Environmental Fake News Detection Utilising Weak Labelling and Machine Learning Techniques

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Environmental Fake News Detection Utilising Weak Labelling and Machine Learning Techniques

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Abstract

The rapid growth of digital media has significantly increased the spread of misinformation, particularly in critical domains such as environmental and disaster-related news. This study presents the design and evaluation of a comparative fake news detection framework that integrates weak labelling, machine learning, deep learning, and transformer-based approaches. Due to the lack of manually annotated datasets in this domain, a rule-based weak labelling strategy was implemented, combining source credibility and domain-specific keyword patterns. The keyword sets were customised with the assistance of AI-based tools to capture linguistic indicators associated with both reliable and misleading content.

The dataset was pre-processed through cleaning, normalization, and feature engineering, including the construction of a combined textual representation. Both textual features (TF-IDF, n-grams, and contextual embeddings) and credibility-based features (source reliability and keyword indicators) were incorporated. Four models were implemented and evaluated: Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa.

The results demonstrate that all models were capable of classifying news articles using weakly labelled data, with transformer-based models achieving slightly better performance due to their contextual understanding. Traditional machine learning models provided competitive baseline results with greater interpretability and lower computational cost. The study also highlights the importance of combining textual and credibility features to improve classification accuracy.

Overall, the research confirms that weak labelling is a practical and scalable approach for domain-specific fake news detection, and that comparative evaluation of different modelling techniques provides valuable insights into their strengths and limitations. The findings contribute to the development of effective and interpretable misinformation detection systems in environmental and disaster communication.

Acronyms

Acronym	Full Form
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
DL	Deep Learning
TF-IDF	Term Frequency – Inverse Document Frequency
LR	Logistic Regression
XGBoost	Extreme Gradient Boosting
BiLSTM	Bidirectional Long Short-Term Memory
LSTM	Long Short-Term Memory
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Approach
DistilRoBERTa	Distilled Version of RoBERTa
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
PR-AUC	Precision-Recall Area Under Curve
MCC	Matthews Correlation Coefficient
EDA	Exploratory Data Analysis
API	Application Programming Interface

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1 Introduction

The rapid growth of digital media has made information widely accessible but also increased the spread of false or misleading content across websites, blogs, and social media. Fake news is a major concern as it can influence public opinion, create confusion, and reduce trust in reliable sources. In critical domains such as environmental and disaster-related news, misinformation can be especially harmful, affecting public awareness, emergency response, and decision-making.

Environmental news covers sensitive topics like climate change, natural disasters, and public safety, where inaccurate information can cause panic, undermine scientific trust, and mislead communities. This highlights the need for automated methods to identify and verify reliable information.

Machine learning and natural language processing offer effective computational solutions. Traditional models like Logistic Regression and XGBoost capture textual patterns, while deep learning and transformer-based models such as BiLSTM and DistilRoBERTa better understand context and semantics.

This study develops a fake news detection framework for environmental and disaster news, incorporating customised credibility features based on domain-specific keywords and source characteristics. Multiple models are trained and compared to evaluate performance, while also analysing interpretability and credibility signals. The research contributes to misinformation detection by focusing on the critical domain of environmental and disaster communication.

1.1 Purpose of this Study

The purpose of this study is to develop and evaluate an automated fake news detection system for environmental and disaster-related news using machine learning, deep learning, and transformer-based methods. The study aims to explore whether computational techniques can effectively identify misleading or unreliable news content in this specific domain.

Another purpose of the research is to compare different modelling approaches in order to understand their strengths and limitations. The study does not focus on a single algorithm; instead, it investigates several models ranging from simple baseline approaches to advanced transformer models. This comparison helps determine which method offers the best balance between performance, interpretability, and practicality.

In addition, the study aims to examine how rule-based weak labelling and credibility-oriented features can support fake news classification when a fully manually labelled dataset is not available.

1.2 Background of the Study

The expansion of digital media has significantly improved access to information while simultaneously increasing the spread of misinformation. Online platforms enable rapid dissemination of news content, making it difficult to verify the accuracy and reliability of information.

This challenge is particularly critical in the context of environmental and disaster-related news, where inaccurate reporting can misinform the public, undermine trust in credible sources, and influence decision-making during sensitive situations. Ensuring the credibility of information in this domain is therefore essential.

To address this issue, computational techniques based on machine learning and natural language processing (NLP) have been widely adopted. Traditional models such as Logistic Regression and XGBoost are effective in identifying textual patterns, while advanced approaches, including BiLSTM and transformer-based models like DistilRoBERTa, provide deeper contextual understanding.

This study develops a domain-specific fake news detection framework for environmental and disaster-related news by integrating weak labelling with machine learning, deep learning, and transformer-based approaches, supported by customised keyword-based features.

1.3 Research Question, Aim and Objectives

The aim of this research is to design, implement, and evaluate a comparative fake news detection framework for environmental and disaster-related news using weak labelling, machine learning, deep learning, and transformer-based models, with the purpose of assessing their classification performance, interpretability, and computational efficiency.

The main research question of this study is:

a) **"Can machine learning and transformer-based models classify environmental and disaster-related news as real or fake using a weakly labelled dataset?"**

To support this main question, the following Objectives are considered:

a. Can a rule-based weak labelling approach be used for generating Real and Fake labels in environmental and disaster-related news data?
b. Which textual and credibility-related features are most useful for distinguishing real news from fake news?

c. Is there a significant variation between the compared between general machine learning models and transformer-based models?

1.4 What Could This Research Uncover?

This research may uncover several important findings related to misinformation detection in environmental and disaster news.

First, it may reveal whether fake and real news articles in this domain contain identifiable linguistic and credibility-related patterns that can be learned automatically by machine learning models. For example, fake articles may use more sensational language, fewer scientific references, or less credible sources.

Second, the research may show how effective weak labelling is as a practical solution when manually annotated data is limited. This is important as many real-world datasets are incomplete or unlabelled, and weak supervision can provide a cost-effective alternative.

Third, the study could uncover which modelling approach is most suitable for this task. Traditional machine learning models may perform well with lower computational cost, while deep learning and transformer models may achieve stronger contextual understanding. Comparing these methods can help identify which technique is most appropriate for academic and practical use.

Finally, the research may provide insights into the broader challenge of domain-specific fake news detection. Much existing work focuses on general misinformation, but the proposed framework may show that environmental and disaster news requires its own tailored features, labels, and evaluation strategies.

1.5 Importance of This Research

This research is important because misinformation in environmental and disaster contexts can have serious real-world consequences. False reports about natural disasters, climate change, public safety warnings, or environmental policies can mislead individuals, increase fear, and reduce trust in authoritative information sources. In some situations, misinformation may even delay response actions or encourage unsafe decisions.

The study is also important from a technical perspective. Fake news detection remains a challenging problem because language is complex, deceptive content evolves over time, and labelled datasets are often limited. By combining weak labelling, feature engineering, traditional machine learning, and transformer-based methods, this research explores a practical and scalable way to address the problem. Academically, the thesis contributes to the field by focusing on a specific domain rather

than treating fake news as a general problem. This domain-focused approach may provide more meaningful findings because environmental and disaster communication has unique vocabulary, credibility indicators, and risks. The results may be useful for future researchers, developers, journalists, and policy stakeholders interested in misinformation monitoring.

1.6 Scope of the Study

The proposed framework focuses on detecting fake news in environmental and disaster-related articles using text fields (title, content) and selected metadata (source, author, date). The dataset is weakly labelled using rule-based methods based on source credibility and keyword patterns, with uncertain cases excluded.

The scope includes preprocessing, feature engineering, model development, and evaluation using models such as Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa. Performance is assessed using standard metrics including accuracy, precision, recall, F1-score, MCC, ROC-AUC, PR-AUC, and log loss.

The study is limited to English text data and does not cover multimedia misinformation, multilingual detection, or real-time deployment, focusing instead on a research prototype for model comparison. In addition, the study does not claim to determine absolute truth in news reporting. Since the labelling process relies partly on heuristic rules, the results must be interpreted within the limitations of weak supervision. Therefore, the scope of this thesis is to evaluate a comparative fake news detection framework within a specific domain and dataset, rather than to create a universally applicable misinformation detection system.

1.7 Summary

This chapter introduced fake news detection in environmental and disaster-related news, highlighting its background, motivation, and associated risks. It outlined the research aim, objectives, key questions, and scope, while also explaining the significance of the study.

Additionally, the chapter described the overall structure of the thesis and how each chapter contributes to the research. It establishes a clear foundation by defining the problem and guiding the direction of the study.

The next chapter presents a review of relevant literature on fake news detection, machine learning, deep learning, transformer-based models, weak labelling, and credibility analysis, providing the theoretical basis for the proposed work.

2 Literature Review

2.1 Introduction

Fake news has become a major issue in the digital age due to the rapid spread of information across websites, blogs, and social media. Unlike traditional media, online platforms enable misleading content to reach large audiences quickly. As a result, fake news detection has gained significant attention in fields such as machine learning, natural language processing, and data science.

The literature shows that fake news detection is not only about checking whether a statement is true or false. Researchers also study language patterns, writing style, source credibility, and the way information is framed. Some studies focus on traditional machine learning methods, while others explore deep learning and transformer-based models. At the same time, researchers continue to discuss the challenges of limited labelled data, changing language patterns, and the need for reliable evaluation methods. Mahmoud et al. (2025) explain that NLP and machine learning techniques are now widely used in fake news classification, while Prachi et al. (2022) show that different algorithms can produce different levels of performance depending on the representation of the text and the model used.

Another important point in the literature is that fake news does not affect all domains in the same way. Some studies focus mainly on political misinformation, while others examine health, social media, and online communication more broadly. Shushkevich et al. (2024) discusses fake news detection in online platforms and highlights the growing challenge of identifying misinformation in fast-moving digital environments. This broader context is useful because it shows that fake news detection is not only a technical task, but also a social and communication problem.

This chapter reviews the main ideas and methods already discussed in the literature. It explains the concept of fake news, discusses its characteristics, and reviews the use of NLP and machine learning in fake news detection. It also looks at deep learning and transformer-based models, credibility-based signals, and common evaluation methods. The chapter ends by identifying a research gap that supports the need for further study in this area.

2.2 Related Works

The problem of fake news detection has received significant attention in recent years due to the rapid growth of online platforms and digital media. Researchers have explored a variety of computational approaches, including natural language processing (NLP), machine learning, and deep learning techniques, to identify and classify misleading information. This section reviews key studies relevant to this research and highlights their contributions, methodologies, and limitations.

Abdulrahman et al. (2020) proposed a fake news detection framework that integrates NLP techniques with machine learning algorithms. Their study emphasizes the importance of preprocessing, feature extraction, and text representation methods such as TF-IDF. They demonstrate that traditional machine learning models can achieve strong performance when combined with effective feature engineering. However, their work primarily focuses on general fake news datasets and does not address domain-specific challenges such as environmental or disaster-related misinformation.

Shushkevich et al. (2024) investigates fake news detection in online platforms, highlighting the role of digital environments in the rapid spread of misinformation. The study emphasises that fake news detection is not only a technical classification problem but also a broader communication challenge influenced by user behaviour, platform dynamics, and information dissemination patterns. While the study provides valuable insights into the online context of misinformation, it focuses less on detailed model comparison and implementation.

Prachi et al. (2022) presents a comparative analysis of machine learning and NLP-based techniques for fake news detection. Their research evaluates different algorithms and demonstrates that model performance depends heavily on text representation and preprocessing methods. The study highlights the effectiveness of algorithms such as Logistic Regression and other classifiers but does not explore advanced transformer-based models in depth.

In addition to these studies, recent research (Rahman et al., 2022) has increasingly explored deep learning and transformer-based approaches, such as LSTM and BERT-based models, which are capable of capturing contextual and semantic relationships within text. These models have shown improved performance in complex classification tasks, particularly where subtle language patterns are involved.

2.3 Fake News: Definition, Concepts and Characteristics

Fake news refers to false or misleading information presented in a news-like format to appear credible, often to attract attention or influence opinions. Because it closely resembles legitimate journalism in structure and language, it is difficult to detect, highlighting the need for automated detection methods.

Researchers, Hemant et al. (2021) also point out that fake news is not always completely invented. In some cases, it may include partial truths, missing context, exaggerated statements, or misleading headlines. For this reason, fake news detection is more difficult than simply identifying obviously false claims. A text may contain real facts but still be misleading in how those facts are presented. This view is supported in recent work, where fake news is treated as a broader credibility and classification problem rather than only a fact-checking problem.

Hemant et al. (2021) describe fake news detection as a task that combines text processing and machine learning in order to classify news content as reliable or unreliable. Their discussion shows that current research treats fake news as both a language problem and a classification problem, which is why NLP techniques are so commonly used in this field.

Shushkevich et al. (2024) emphasizes that such content spreads rapidly in online environments, making detection more challenging.

Another common characteristic is the absence of strong supporting evidence. Reliable news articles are more likely to include verifiable facts, references to official statements, precise figures, and identifiable sources. In contrast, fake news may rely on vague claims, anonymous sources, unsupported statements, or loosely connected assertions. The presentation of information may appear confident and persuasive, but closer examination often reveals a lack of evidence.

Figure 2.1 illustrates the key differences between real news and fake news across several important dimensions such as credibility, sourcing, language style, intent, and verifiability.

Busioc et al. (2020) note that NLP and ML approaches can capture many of these characteristics through text analysis. Their work supports the idea that lexical patterns, feature extraction, and source-related signals are useful for distinguishing fake news from legitimate reporting. Shakya and Jain (2019) also show that machine learning models can learn such differences when trained on processed textual data.

Figure 2.1:Comparison of Real and Fake News Characteristics

2.4 Natural Language Processing in Fake News Detection

Natural language processing plays a central role in fake news detection because most fake news studies analyse written text. NLP helps convert raw news content into a form that can be processed by machine learning models. Researchers use NLP to clean text, extract features, identify meaningful patterns, and prepare data for classification. Without these steps, it would be difficult for computational systems to learn useful differences between reliable and unreliable news.

Madani et al. (2023) place NLP at the centre of fake news detection and explain that text representation and preprocessing are essential for successful classification. Their work supports the general view in the literature that fake news detection begins with effective language processing.

2.4.1 Text Preprocessing Techniques

Feature extraction converts text into numerical representations for model use. Early fake news studies relied on lexical features such as word frequency, n-grams, and bag-of-words, which capture differences in vocabulary and writing style.

Text preprocessing is a key initial step, typically involving lowercasing, removal of punctuation, stop words, URLs, and extra whitespace. These steps reduce noise and improve consistency, which is especially important for online news data, Hadeer et al. (2017).

Researchers differ slightly in how much preprocessing they apply. Some use simple cleaning techniques, while others apply stemming or lemmatization. Even with these differences, the literature clearly shows that preprocessing improves text quality and supports better feature extraction. Mahmoud et al. (2025) and Prachi et al. (2022) both reflect this common pattern by emphasizing preprocessing as an important early stage in their fake news detection workflows.

2.4.2 Feature Extraction Methods

Feature extraction converts text into numerical form so that machine learning models can analyse it. Existing research uses methods such as bag-of-words, n-grams, TF-IDF, and embeddings, Mahmoud et al. (2025). These methods help capture word frequency, word importance, and sometimes contextual meaning.

The literature shows that feature extraction is one of the most important parts of fake news detection because model performance depends heavily on how the text is represented. A weak representation may reduce classification accuracy, while a strong representation can improve even relatively simple models. This is one reason why many studies compare different feature extraction approaches alongside different classifiers.

2.4.3 TF-IDF in Fake News Detection

TF-IDF is a widely used method for text representation that highlights words important to a document by combining term frequency with inverse document frequency, reducing the impact of common words.

It is particularly effective with interpretable models like Logistic Regression, where feature weights can reveal which terms are associated with each class, making it valuable for research focused on interpretability.

Mahmoud et al. (2025) discuss TF-IDF as one of the important methods for representing textual data in fake news classification. Their work supports the broader literature, where TF-IDF remains a strong baseline technique because it offers a good balance between simplicity and effectiveness. Prachi et al. (2022) also examines NLP-based methods in machine learning workflows, reinforcing the importance of proper textual representation in model performance.

2.5 Machine Learning Approaches

Machine learning approaches play a key role in fake news detection, as they enable automated classification of news content based on textual and credibility features. Common models include Logistic Regression, Naïve Bayes, Support Vector Machine, Decision Tree, Random Forest, and boosting methods. These models are widely used due to their simplicity, efficiency, and suitability for experimental comparison.

Jain et al. (2019) highlight that traditional machine learning methods remain important due to their strong baseline performance and practicality. Similarly, Shushkevich et al. (2024) emphasizes the need for scalable and efficient models in online environments, reinforcing the relevance of machine learning techniques.

2.5.1 Traditional Machine Learning Models

Traditional machine learning models are widely used in fake news detection due to their low computational cost and ease of implementation. These models rely on structured feature representations such as TF-IDF to capture patterns in textual data and distinguish between real and fake news.

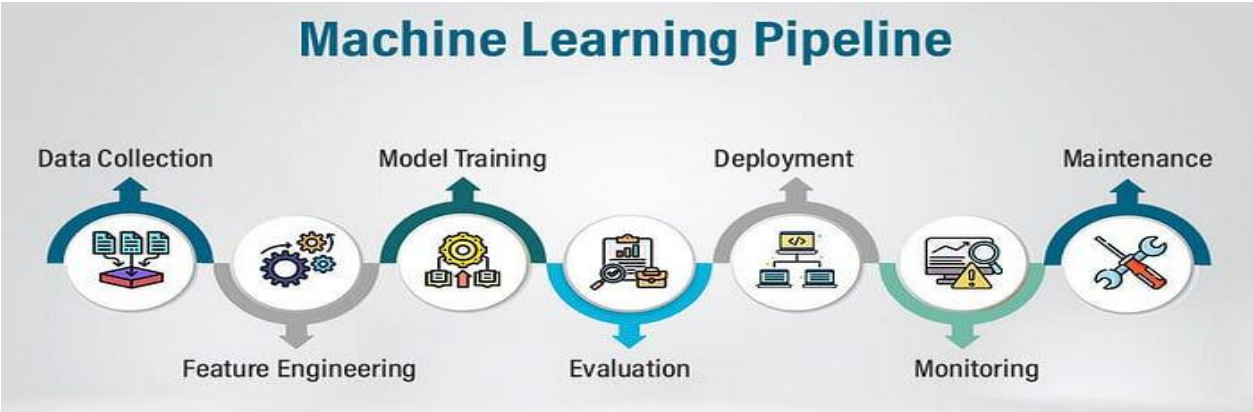

Figure 2.2:Machine Learning Pipeline

Figure 2.2 presents the standard pipeline followed in traditional machine learning models, particularly for tasks such as text classification (e.g., fake news detection). The figure outlines a sequence of well-defined stages that transform raw data into predictive insights.

Logistic Regression is a widely used algorithm for binary classification tasks. It estimates the probability that a news article belongs to a specific class based on weighted input features. When combined with TF-IDF, it can effectively identify patterns in word usage associated with fake or real news. A key advantage of Logistic Regression is its interpretability. Model coefficients indicate

the importance of each feature, allowing researchers to understand which terms influence predictions. This makes it particularly useful in fake news detection, where transparency is important. Hossain and Rahman (2022) and Mahmoud (2025) both identify Logistic Regression as a reliable and effective baseline model.

XGBoost is an advanced ensemble learning algorithm based on gradient boosting. It builds multiple decision trees sequentially, allowing it to capture complex and non-linear relationships between features. Compared to linear models, XGBoost can better model interactions between features and is particularly useful when working with engineered features such as credibility indicators. However, its performance depends on careful parameter tuning and may vary depending on the dataset. Mahmoud and Ibrahim (2025) highlight the growing use of boosting-based methods in fake news detection, reflecting a broader trend towards more powerful machine learning techniques.

2.6 Deep learning and transformer-based approaches

Deep learning and transformer-based approaches have become increasingly important in fake news detection due to their ability to capture contextual and semantic relationships in text. Unlike traditional machine learning models, which rely on manually engineered features, these approaches learn representations automatically from data, making them more effective for complex language patterns. Recent studies highlight the growing use of deep learning models such as LSTM and BiLSTM, as well as transformer-based models such as BERT and RoBERTa, in text classification tasks. Chauhan and Palivela (2021) note that these methods improve performance by capturing deeper linguistic features, while Lohia and Tilak (2018) demonstrate their effectiveness in fake news detection tasks.

2.6.1 Deep Learning Models

Deep learning models, particularly LSTM and BiLSTM, are widely applied in fake news detection due to their ability to process text as sequential data and capture contextual relationships between words. Unlike traditional machine learning approaches, these models do not rely heavily on manual feature extraction; instead, they automatically learn meaningful representations from raw text.

LSTM models are designed to handle long-term dependencies by using memory cells and gating mechanisms that control the flow of information. This allows them to retain important context from earlier parts of a sentence while ignoring irrelevant details. Devlin et al. (2019) As a result, they are well-suited for understanding complex language patterns.

BiLSTM models extend this capability by processing sequences in both forward and backward directions. This dual perspective enables the model to capture both preceding and succeeding context, leading to improved accuracy in tasks where word meaning depends on surrounding text. Overall, deep learning models provide more robust and context-aware predictions, making them highly effective for identifying subtle patterns in fake news content.

2.6.2 Transformer-Based Models

Transformer-based models have significantly advanced natural language processing by introducing self-attention mechanisms that capture relationships between all words in a text. Models such as BERT and RoBERTa provide strong contextual understanding, making them highly effective for fake news detection. DistilRoBERTa, a lightweight version of RoBERTa, offers a balance between performance and efficiency, making it suitable for practical applications.

For these reasons, the literature increasingly emphasizes comparative studies rather than assuming that more complex models are always preferable. The suitability of a model depends not only on accuracy, but also on transparency, efficiency, and alignment with the problem context.

2.7 Weak Labelling and Credibility Signals in Fake News Detection

Weak labelling has gained increasing attention in machine learning research as a practical solution to the scarcity of high-quality annotated data. In the context of fake news detection, manual annotation is particularly challenging, as it often requires extensive source verification, contextual understanding, and fact-checking. To address this limitation, researchers have explored weak labelling techniques that rely on heuristic rules, keyword matching, source-based credibility lists, and metadata signals to generate approximate labels (Saad et al., 2017; Lohia et al., 2018). Although such labels may introduce noise, prior studies demonstrate that they can still support effective supervised learning, especially when gold-standard datasets are unavailable (Umer et al., 2020).

In addition to weak labelling, the literature highlights the importance of credibility-related signals in fake news detection. Source credibility is widely recognised as a key indicator, as information originating from official organisations, scientific institutions, and reputable media outlets is generally considered more reliable (Sliva et al., 2017). Keyword-based approaches are also commonly used, particularly in domain-specific contexts where certain terms may reflect scientific reporting or sensational framing (Rashkin et al., 2017). Furthermore, researchers examine additional credibility indicators such as the presence of numerical evidence, references to experts, balanced

language, and writing style, all of which can contribute to assessing the trustworthiness of content (Yuan et al., 2020).

Overall, existing research demonstrates that weak labelling and credibility signals are closely interconnected in fake news detection. While neither approach provides a complete solution independently, both play a crucial role in scenarios where manually labelled data is limited and where evaluating the reliability of information is essential.

2.8 Fake News Detection in Online Platforms

Shushkevich (2024) provides a useful perspective by focusing on fake news detection in online platforms. This is important because digital platforms are one of the main spaces where misinformation spreads rapidly. The online environment allows content to be created, shared, and reshaped quickly, making manual verification difficult. Studies in this area highlight that fake news detection must be understood not only as a text classification task but also as a broader platform and communication challenge. This broader perspective strengthens the fake news literature because it shows that computational methods are part of a larger information ecosystem. It also helps explain why source credibility and context are important alongside text-based features.

2.9 Evaluation Metrics in Fake News Detection

Evaluation is a very important part of fake news detection research because the quality of a model cannot be judged by one number alone. In classification tasks, especially those involving misleading or harmful content, different kinds of errors may have different consequences. A model may appear to perform well overall, but still fail to identify fake articles correctly or produce too many false alarms. For this reason, the literature consistently recommends the use of multiple evaluation metrics when comparing fake news detection models. Mahmoud et al. (2025) and Prachi et al. (2022) both reflect this broader research practice by discussing or applying comparative evaluation across different NLP and machine learning approaches.

The literature also shows that the choice of evaluation metric depends partly on the goal of the system. If the main concern is to avoid wrongly labelling true news as fake, then precision becomes particularly important. If the goal is to detect as much fake news as possible, then recall may be more important. In other words, different metrics help researchers understand different aspects of model behaviour. This is why comparative fake news detection studies usually report several metrics

together rather than depending on a single score. Prachi et al. (2022), for example, compare different approaches in a way that reflects this wider pattern in the field.

2.9.1 Accuracy, Precision, Recall, and F1-Score

Accuracy measures the overall proportion of correct predictions, but it does not always provide a complete picture of model performance. For this reason, researchers also use precision, recall, and F1-score. Precision shows how many articles predicted as fake are actually fake, while recall shows how many fake articles are correctly detected. F1-score combines both measures and is widely used because it gives a more balanced view of classification quality. These metrics are commonly reported in fake news detection studies to provide clearer comparison across models.

2.9.2 ROC-AUC and PR-AUC

ROC-AUC and PR-AUC are also important in fake news detection because they measure how well a model separates real and fake classes across different threshold settings. ROC-AUC gives a general view of classification ability, while PR-AUC is especially useful when the fake class is of particular interest. The literature increasingly recommends these measures because they provide deeper insight than basic accuracy alone.

2.10 Research Gap

Although fake news detection has been widely studied, the literature still shows some important gaps. First, much of the existing work focuses on general fake news, politics, or broad online misinformation rather than domain-specific areas. Second, many studies compare machine learning and deep learning models, but fewer focus specifically on how such approaches perform in specialized contexts. Third, source credibility and context are often discussed in the literature, but they are not always integrated in the same way across studies.

Another gap is that many papers focus heavily on model performance while giving less attention to readability, interpretability, and the practical value of detection systems in real online environments. The work by Shushkevich (2024) helps show that online fake news detection is a wider communication problem, not only a technical classification task. Mahmoud and Ibrahim (2025) and Prachi (2022) show the strong role of NLP and ML methods, but the literature still leaves room for clearer comparative work and more domain-focused studies.

Another limitation is that many studies rely on datasets that are already manually labelled. These datasets are useful for controlled evaluation, but they may not reflect the practical challenges of working with raw or weakly structured information. In many real-world domains, large manually annotated corpora simply do not exist. This reduces the applicability of some existing findings to domain-specific or resource-limited settings.

Overall, the literature suggests that fake news detection remains an active and evolving area of research, with continuing need for clearer comparison, domain-sensitive analysis, and strong integration of NLP and machine learning methods.

2.11 Summary

This chapter reviewed the main literature related to fake news detection and showed that the field combines concepts from communication studies, natural language processing, and machine learning. It discussed the meaning and characteristics of fake news, the role of misinformation in digital environments, and the ways in which NLP techniques are used to process and represent textual data for classification tasks.

The chapter also examined traditional machine learning methods, deep learning models, and transformer-based approaches that have been applied in fake news detection research. In addition, it highlighted the importance of source credibility, weak labelling, and credibility-related indicators as useful signals in identifying unreliable news content. Finally, the chapter reviewed common evaluation metrics used in previous studies and showed that comparative, multi-metric evaluation is essential for understanding model performance.

Overall, the literature indicates that fake news detection is an active and developing research area. Existing studies have made important progress, but they also reveal continuing challenges related to domain differences, data quality, interpretability, and evaluation. These findings provide the theoretical background for the next chapter, which presents the design and implementation of the system developed for this research.

3 Design and Implementation

This chapter presents the design and implementation of the proposed fake news detection framework for environmental and disaster-related news. It explains how the research problem was translated into a structured computational system and outlines the main stages involved in building the framework. While the earlier chapters introduced the background, research questions, and theoretical context of the study, this chapter focuses on the practical development of the solution. The aim is to show how the dataset, preprocessing techniques, weak labelling process, feature preparation, and classification models were brought together to create a coherent and reproducible research pipeline.

3.1 Introduction

The framework was developed to support the classification of environmental and disaster-related news as either Real or Fake using a weakly labelled dataset. A comparative design was adopted so that multiple modelling approaches could be implemented and evaluated within the same pipeline. This was important because the study does not seek only to build a single fake news detector, but also to examine how different methods perform in this domain. The implementation therefore includes traditional machine learning, deep learning, and transformer-based techniques, allowing the research to assess differences in predictive performance, interpretability, and computational efficiency.

A significant challenge in the design of the system was the absence of a fully manually annotated dataset. To address this limitation, the framework incorporates a weak labelling strategy based on source credibility and domain-specific keyword patterns. This makes the implementation more realistic, as real-world misinformation studies often face limited access to verified labels. In addition to label generation, the framework includes preprocessing steps such as handling missing values, removing duplicate records, cleaning text, and constructing a combined textual representation from article titles and article content. These steps were necessary to ensure the consistency and quality of the input data before model training.

Overall, this chapter provides the technical foundation for the rest of the thesis by describing how the proposed system was designed and implemented in practice. It begins with an overview of the framework and the research methodology, followed by a description of the dataset used in the study. The later sections explain the preprocessing pipeline, weak labelling process, feature engineering, model development, and training setup. Together, these components form the basis of the experimental work presented in the results and evaluation chapter.

3.2 System Overview

The proposed fake news detection system follows a structured, multi-stage pipeline designed to process environmental and disaster-related news data and classify it as real or fake. The overall architecture of the system is illustrated in Figure 3.1.

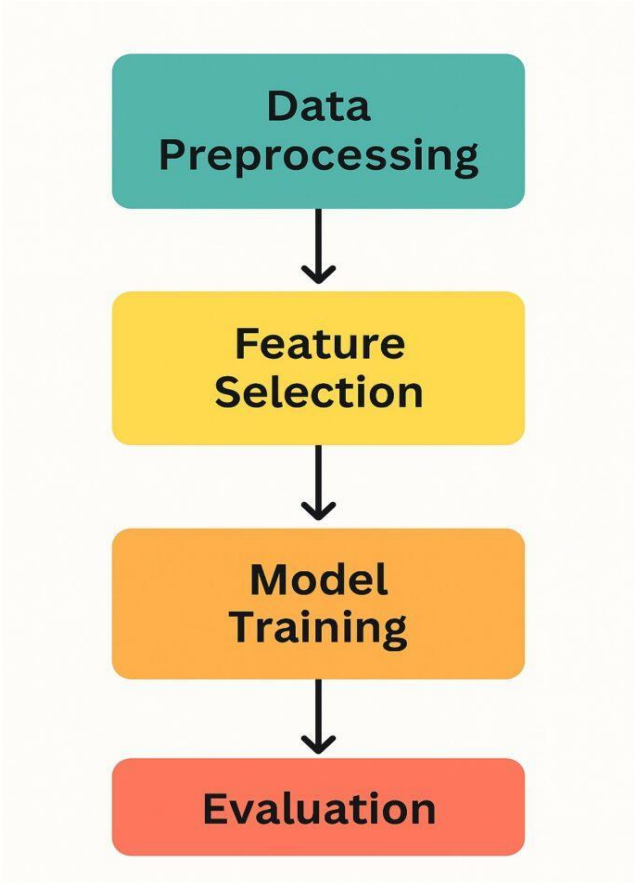

Figure 3.1: Simplified Fake News Detection Pipeline

Figure 3.1 presents a simplified overview of the fake news detection system. The process begins with data collection, where environmental and disaster-related news articles are gathered from various sources. This is followed by data preprocessing, where the text is cleaned, normalised, and prepared for analysis.

In the next stage, multiple models are trained using the processed data to learn patterns associated with real and fake news. Finally, the trained models are evaluated using standard performance metrics to assess their classification effectiveness.

While the simplified diagram provides an overview of the system, a more detailed representation is required to illustrate the individual components and processes involved. The complete system architecture is shown in Figure 3.2.

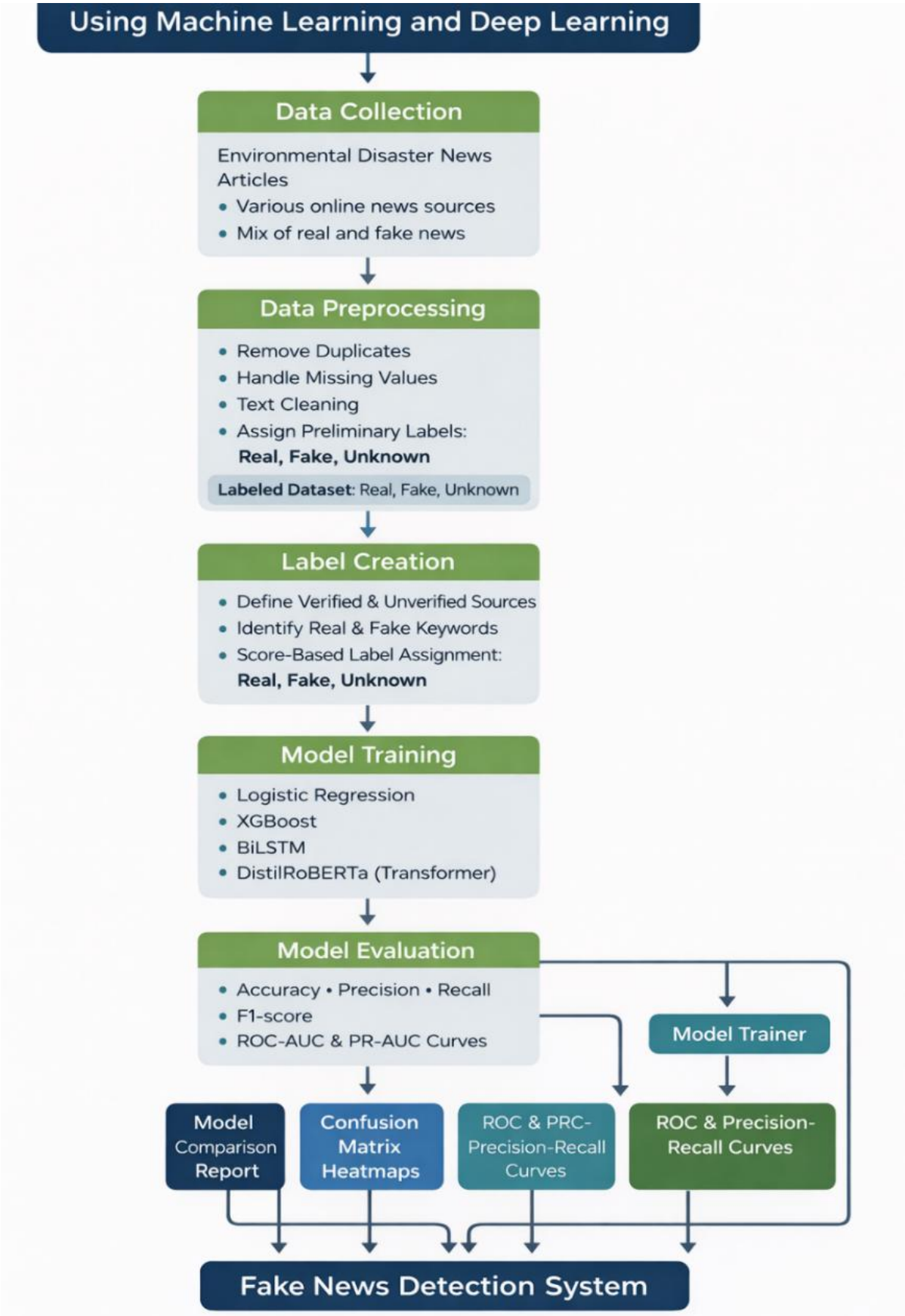

Figure 3.2:Overall System Architecture for Fake News Detection

Figure 3.2 illustrates the complete workflow of the proposed system. The process begins with data collection, followed by data preprocessing, where textual and metadata features are cleaned, normalised, and standardised.

A weak labelling strategy is then applied to assign labels based on source credibility and keyword-based rules. This step enables the creation of a labelled dataset without requiring manual annotation. Articles that cannot be confidently labelled are assigned an Unknown category and excluded from supervised training.

Following this, feature engineering is performed to convert the processed data into numerical representations suitable for machine learning and deep learning models. This includes TF-IDF for traditional models and tokenization for deep learning and transformer-based models.

The system then proceeds to the model training stage, where multiple classification models are implemented, including Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa. This multi-model approach allows for a comprehensive comparison of different learning techniques.

Finally, the models are evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. The results are then compared to determine the most effective approach for fake news detection in this domain.

3.3 Research Implementation

This research adopts a structured experimental methodology to design, implement, and evaluate a fake news detection system for environmental and disaster-related news. The methodology follows a data-driven approach, integrating preprocessing, weak labelling, feature engineering, and comparative model evaluation within a unified framework.

The overall research process is organised into a sequence of interrelated stages, as illustrated in Figure 3.1. Each stage is designed to transform raw textual data into meaningful representations that can be used for classification and analysis.

The first stage involves data collection and preparation, where a dataset of environmental and disaster-related news articles is obtained and prepared for analysis. This includes standardising data formats, removing inconsistencies, and ensuring that relevant textual and metadata features are available for processing.

The second stage focuses on data preprocessing, where raw text is cleaned and normalised to improve data quality. This involves removing duplicate entries, handling missing values, and

applying text cleaning techniques such as lowercasing, removal of noise (e.g., URLs and special characters), and whitespace normalisation. These steps ensure that the data is consistent and suitable for feature extraction.

The third stage introduces a weak labelling strategy, which is used to generate class labels in the absence of fully annotated data. A rule-based approach is applied, incorporating source credibility and domain-specific keyword patterns to assign labels such as Real, Fake, and Unknown. Articles labelled as Unknown are excluded from supervised model training to maintain label reliability.

Following this, feature engineering is performed to transform textual data into numerical representations. For traditional machine learning models, TF-IDF is used to capture term importance and document-level patterns. For deep learning models, tokenization and sequence padding are applied to prepare text for sequential processing. This dual representation approach enables the use of both classical and advanced models within the same framework.

The next stage involves model development and training, where multiple classification models are implemented, including Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa. These models are selected to represent different learning paradigms, allowing for a comprehensive comparison of performance across traditional machine learning, deep learning, and transformer-based approaches.

Finally, the models are assessed in the evaluation stage using a range of performance metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Additional evaluation tools such as confusion matrices and performance curves are used to analyse model behaviour and classification effectiveness.

Overall, this methodology provides a systematic and reproducible framework for investigating fake news detection in a domain-specific context. It enables both performance comparison and interpretability analysis, while addressing the challenge of limited labelled data through the use of weak supervision.

3.4 Dataset Description

The dataset consists of environmental and disaster-related news articles compiled in an Excel file from sources including GitHub, Kaggle, social media, and radio broadcasts. It contains both textual content and associated metadata, providing the basis for fake news classification and credibility analysis. The main fields used in the implementation are title, text, date, source, and author. These attributes were selected because they offer both linguistic information and contextual signals relevant to distinguishing between reliable and misleading news content.

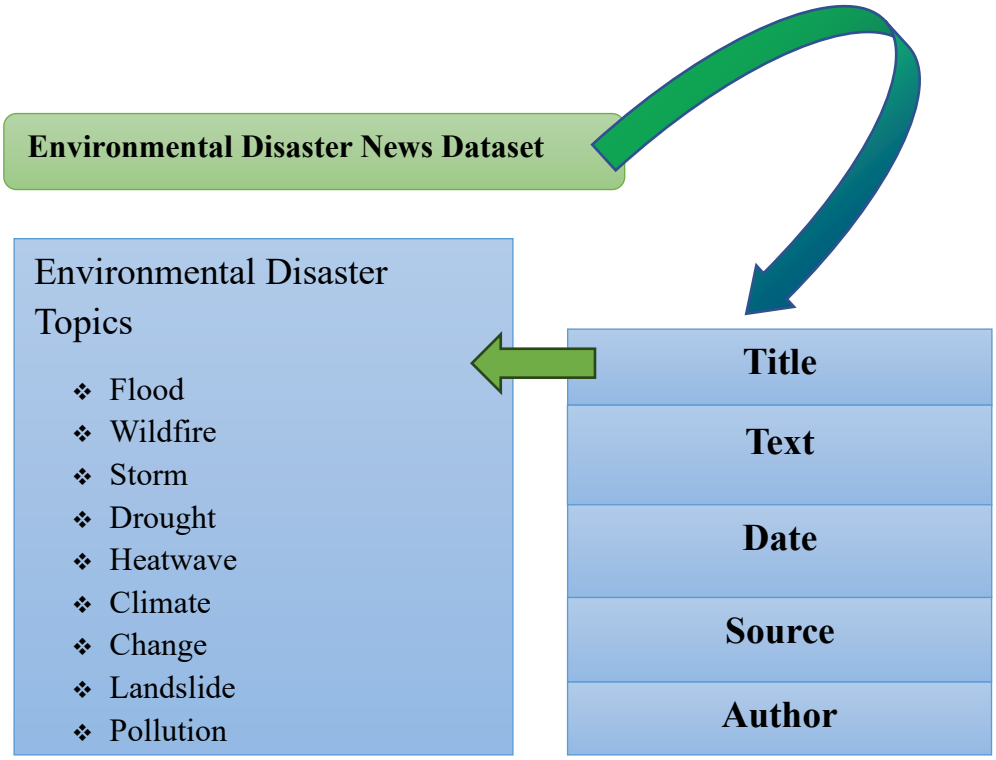

Figure 3.3: Structure of Environmental Disaster News Dataset and Topic Categories

As shown in Figure 3.3, the dataset contains textual features (title and article content) along with metadata such as date, source, and author. It spans topics including floods, wildfires, and climate change. The title captures concise or attention-grabbing claims, while the article text provides detailed context, allowing analysis of both summary- and content-level patterns. The source field is key for weak labelling due to its role in assessing credibility, whereas author and date serve as supplementary metadata.

The main features of the dataset used in this study are summarised in Table 3.1.

Feature	**Description**
Title	Headline of the news article
Text	Full article content
Source	Publisher
Author	Article author
Date	Publication date
Combined Text	Title + Text

Table 3.1:Dataset Features

A notable characteristic of the dataset is that it is domain-specific. Unlike general fake news datasets, this dataset focuses specifically on environmental and disaster communication. This includes articles related to issues such as climate change, natural disasters, environmental risks, public warnings, and crisis-related reporting. The domain-specific nature of the data is important because the vocabulary, source patterns, and credibility indicators in environmental reporting may differ significantly from those found in political, entertainment, or general misinformation datasets.

Another important feature of the dataset is that it is not fully manually annotated with binary labels. Instead, the classification labels used in this study are generated through a weak labelling process. Articles are assigned preliminary labels based on rule-based scoring that considers the credibility of the source and the presence of domain-relevant linguistic indicators. As a result, the final modelling dataset is derived from a larger collection of articles after preprocessing and label assignment. Only articles labelled as Real or Fake are used for supervised model training, while uncertain cases are excluded.

The dataset therefore supports the objectives of the study in two important ways. First, it provides realistic domain-specific news content for experimentation. Second, it enables the investigation of fake news detection in a setting where weak supervision is necessary due to the absence of fully verified manual labels. This makes the dataset highly relevant to the research aim and appropriate for the comparative framework developed in this thesis.

3.5 Data Preprocessing

Data preprocessing is a critical stage in the development of the fake news detection system, as it ensures that the raw dataset is clean, consistent, and suitable for analysis. Environmental and disaster-related news data collected from online sources often contains noise, missing values, duplicates, and inconsistencies that can negatively impact model performance if not addressed properly.

The preprocessing pipeline implemented in this study follows a structured approach that includes handling missing values, removing duplicate records, cleaning and normalizing text, constructing a combined textual representation, and standardizing metadata fields. These steps are designed to improve data quality, reduce noise, and enhance the effectiveness of feature extraction and model training.

As illustrated in Figure 3.4, the preprocessing pipeline transforms the raw environmental news dataset into a structured and clean dataset through multiple stages, including missing value handling, duplicate removal, text cleaning, feature preparation, and label assignment.

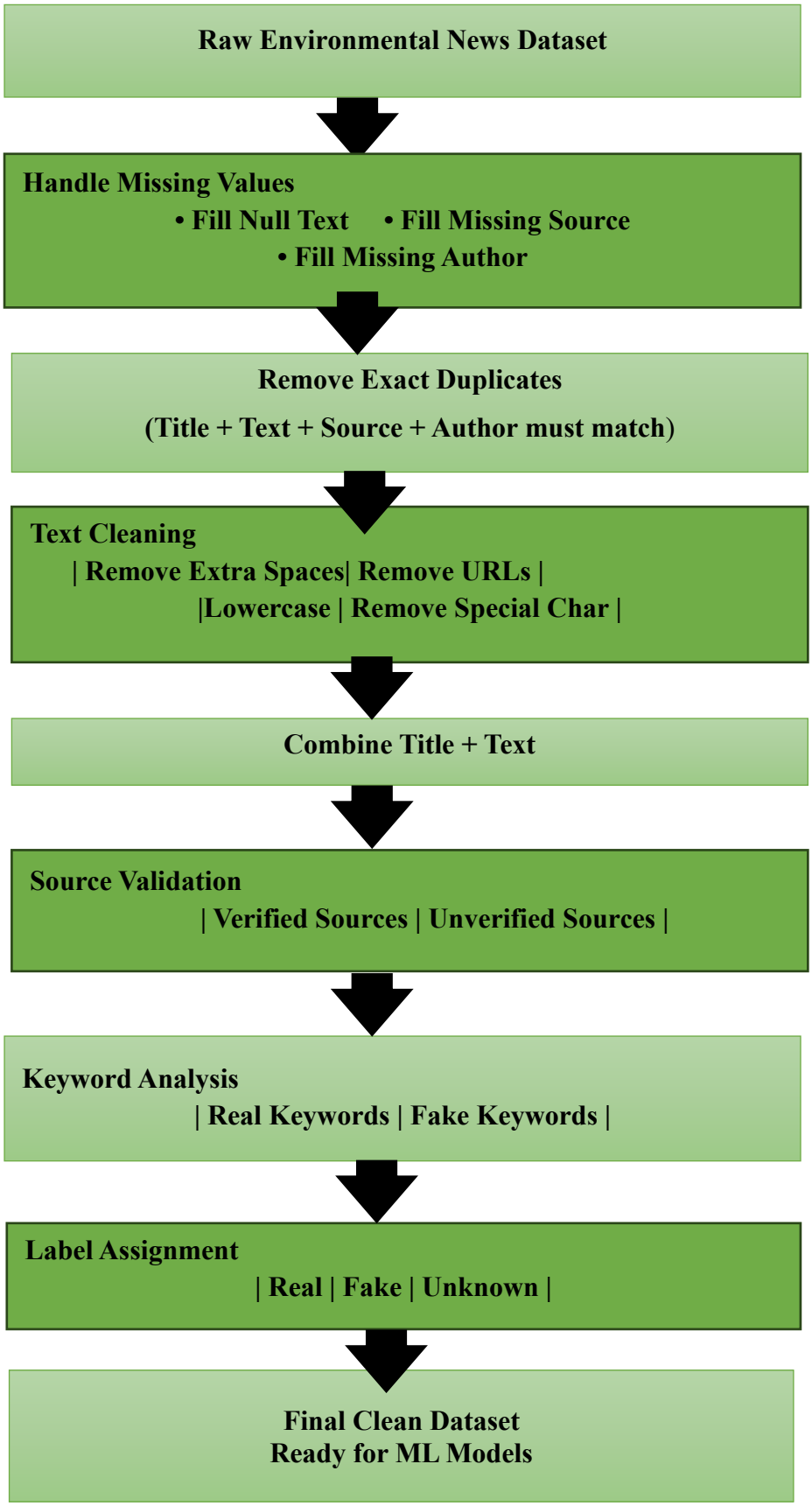

Figure 3.4: Data Preparation Pipeline

This structured pipeline ensures that both textual and credibility-based signals are incorporated into the dataset, improving the effectiveness of the subsequent machine learning and deep learning models.

3.5.1 Handling Missing Values

Missing values were systematically addressed to ensure data consistency. Records lacking essential text fields were removed or completed where possible. For metadata such as source and author, missing entries were replaced with placeholder values (e.g., “unknown”) to preserve dataset structure. This reduces bias and improves model reliability.

3.5.2 Removing Duplicates

Duplicate records—identified by identical title, text, source, and author fields—were removed. This prevents overfitting, reduces bias, and ensures each entry represents unique information, leading to more reliable model evaluation.

3.5.3 Text Cleaning and Normalization

Text data were cleaned and standardized by converting to lowercase, removing URLs, special characters, punctuation, and extra whitespace. These steps reduce noise and variability, enabling models to focus on meaningful linguistic patterns and improving feature extraction. The part is illustrated below for reference.

```
def clean_text(text):

    text = str(text).lower()  text = re.sub(r'http\\S+|www\\S+|https\\S+', '', text)

    text = re.sub(r'<.*?>', '', text)  text = re.sub(r'[^a-zA-Z\\s]', ' ', text)

    text = re.sub(r'\\s+', ' ', text).strip()

    return text

df['clean_text'] = df['combined_text'].apply(clean_text) df['clean_source'] = df['source'].str.lower().str.strip()

df['clean_author'] = df['author'].str.lower().str.strip()

print(df[['title', 'source', 'clean_text']].head())
```

3.5.4 Combined Text Construction

The title and article content were merged into a single text field to capture both headline-level and detailed context. This combined representation enhances the model’s ability to detect patterns such as bias or sensationalism and serves as the primary input for training.

3.6 Weak Labelling Strategy

In many real-world fake news detection tasks, obtaining a fully manually labelled dataset is difficult due to the time, cost, and expertise required for verification. This challenge is particularly relevant in environmental and disaster-related news, where determining the reliability of information often requires domain knowledge and access to trusted sources.

To address this limitation, this study adopts a weak labelling strategy, where labels are generated automatically using heuristic rules instead of manual annotation. The approach combines source credibility and keyword-based signals to assign approximate labels to news articles. Although weak labelling may introduce some noise, it provides a practical and scalable solution for building supervised learning models when labelled data is limited.

The weak labelling process implemented in this study consists of three main components: source-based labelling, keyword-based labelling, and a scoring mechanism that integrates these signals to produce final labels.

3.6.1 Source-Based Labelling

Source credibility is one of the most important indicators of news reliability. In this study, each article is evaluated based on the credibility of its source.

Sources were categorized into two groups:

- Verified Sources: Established and reputable news organisations, scientific institutions, and official agencies. Articles from these sources are more likely to be reliable and are assigned a positive credibility signal.
- Unverified Sources: Unknown, less transparent, or potentially unreliable websites. Articles from these sources are considered more likely to contain misleading or false information and are assigned a negative credibility signal.

This classification allows the system to incorporate external trust signals into the labelling process. By using source credibility as a feature, the model can learn patterns associated with reliable and unreliable reporting.

3.6.2 Keyword-Based Labelling

In the present work, a weak labelling approach was adopted due to the absence of a fully manually annotated dataset. As part of this process, the dataset was customised using a keyword-based labelling strategy. Domain-specific keywords relevant to environmental and disaster-related news were identified and refined with the assistance of artificial intelligence. These keywords were used

to capture linguistic patterns associated with both credible reporting (e.g., scientific terminology, factual descriptions) and misleading content (e.g., sensational or emotionally charged language).

The identified keyword sets were then integrated with source-based credibility signals to support the labelling process. Each article was analysed for the presence and frequency of these keywords, and a scoring mechanism was applied to assign labels such as *Real*, *Fake*, or *Unknown*. This approach enabled the creation of a structured dataset suitable for supervised learning, while maintaining domain relevance and reducing the need for manual annotation.

This customised approach enhances the weak labelling process by incorporating domain-relevant linguistic signals, improving the quality and reliability of the generated labels.

Two categories of keywords were defined:

- Real Keywords: Terms associated with scientific reporting, factual descriptions, and neutral language (e.g., references to data, official reports, or environmental terminology).
- Fake Keywords: Terms associated with sensationalism, exaggeration, or emotionally charged language (e.g., alarming phrases, dramatic expressions, or misleading claims).

Each article is analysed for the presence of these keywords. The frequency and type of keywords provide additional signals that help distinguish between reliable and unreliable content.

Keyword-based labelling is particularly useful in domain-specific datasets because environmental news often contains recurring terminology that can indicate credibility or bias.

To combine the signals from source-based and keyword-based labelling, a scoring mechanism was implemented. This mechanism assigns a numerical score to each article based on its credibility indicators.

The scoring process works as follows:

- A positive score is assigned for indicators associated with reliable content (e.g., verified sources, real keywords).
- A negative score is assigned for indicators associated with potentially fake content (e.g., unverified sources, fake keywords).

The overall score for each article is calculated by aggregating these signals. This allows the system to balance multiple factors rather than relying on a single rule.

The scoring mechanism improves the robustness of the weak labelling process by reducing the impact of individual noisy signals. For example, an article from an unverified source may still be classified as reliable if it contains factual indicators.

```
verified_sources = [
    'reuters', 'bbc', 'national geographic', 'nasa', 'noaa', 'unep','RTÉ',
    'ipcc', 'who', 'un', 'world bank','eu','gove','epa','Al Jazeera Climate','Weather Channel','BBC Radio Ulster','RTA Radio','RTÉ Radio'
]

unverified_sources = [
    'facebook', 'whatsapp', 'telegram', 'unknown blog', 'blogspot','Twitter',
    'wordpress', 'anonymous', 'social media'
]

real_keywords = [
    'official report', 'government report', 'scientific study',
    'climate report', 'meteorological department', 'satellite data',
    'confirmed', 'verified', 'researchers', 'evidence',
    'environment agency', 'disaster management authority',
    'warning issued', 'evacuation order', 'according to experts','Infrastructure Desk','Public Services','Science Desk'
]

fake_keywords = [
    'hoax', 'conspiracy', 'rumor', 'shocking truth', 'unbelievable',
    'secret agenda', 'fake alert', 'viral claim', 'click here',
    'unverified', 'anonymous source', 'without proof',
    'miracle cure', 'hidden truth', 'exposed', 'leaked'
]
```

Figure 3.5: Implementation of Source-Based and Keyword-Based Lists for Weak Labelling

The keyword lists and source credibility categories were implemented programmatically in Python. A sample of the predefined verified sources, unverified sources, and keyword sets used in the weak labelling process is shown in Figure 3.5.

3.6.3 Final Label Assignment

Based on the calculated score, each article is assigned a final label. The following categories are used:

- Real: Articles with a strong positive credibility score
- Fake: Articles with a strong negative credibility score
- Unknown: Articles with ambiguous or neutral scores

Articles labelled as Unknown are excluded from the final training dataset to ensure higher label quality and improve model performance.

This final labelling step produces a structured dataset suitable for supervised learning. While the labels are not manually verified, the combination of source credibility, keyword analysis, and scoring provides a reasonable approximation of ground truth.

```
for src in verified_sources:
    if src in source:
        real_score += 3
        reasons.append(f"verified_source:{src}")
```

```
    for src in unverified_sources:
        if src in source:
            fake_score += 3
            reasons.append(f"unverified_source:{src}")
    # Keyword-based scoring
    real_kw_count, real_matched = count_keyword_matches(text, real_keywords)
    fake_kw_count, fake_matched = count_keyword_matches(text, fake_keywords)
    real_score += real_kw_count    fake_score += fake_kw_count
    if real_matched:
        reasons.append("real_keywords:" + ",".join(real_matched))
    if fake_matched:
        reasons.append("fake_keywords:" + ",".join(fake_matched))
    # Final label
    if real_score > fake_score:
        label = "Real"
    elif fake_score > real_score:
        label = "Fake"
    else:
        label = "Unknown"
```

The above code was implemented to perform the final labelling of the dataset.

Overall, the weak labelling strategy enables the construction of a usable dataset in situations where manual annotation is not feasible. It also reflects real-world conditions, where automated systems must often operate with imperfect or partially labelled data.

3.7 Feature Engineering

Feature engineering converts pre-processed data into numerical forms for modelling. This study uses both textual and credibility-based features for environmental and disaster news. Textual features are derived from combined title and content using TF-IDF and n-grams for traditional models, while deep learning models use tokenized sequences to capture context. Custom credibility features are also introduced, including source reliability, keyword indicators, and metadata consistency. A credibility score is computed by combining positive (reliable) and negative (suspicious) signals, and is used to improve classification and support weak labelling.

3.8 Tools and Technologies Used

This study is implemented in Python using libraries for data processing and modelling. Pandas and NumPy handle data operations, while Scikit-learn and XGBoost are used for machine learning and TF-IDF feature extraction. For deep learning, TensorFlow/Keras are used to build the BiLSTM

model, and Hugging Face Transformers enables the use of DistilRoBERTa for contextual classification. Matplotlib and Seaborn are used for result visualization. A summary of the tools and technologies used in this study and their respective purposes is presented in Table 3.2.

Tool / Library	Purpose
Python	Programming language
Pandas	Data handling and preprocessing
NumPy	Numerical computations
Scikit-learn	Machine learning and TF-IDF
XGBoost	Gradient boosting model
TensorFlow / Keras	Deep learning (BiLSTM)
Hugging Face Transformers	DistilRoBERTa model
Matplotlib / Seaborn	Data visualisation

Table 3.2: Tools and Technologies Used

3.9 Training Setup

To evaluate the performance of the proposed models, the dataset was divided into training and testing sets. This ensures that the models are trained on one portion of the data and evaluated on unseen data, allowing for a fair assessment of their performance.

For traditional machine learning models such as Logistic Regression and XGBoost, TF-IDF features were used as input. For deep learning and transformer models, the text was tokenized and converted into numerical sequences. Padding was applied where necessary to ensure consistent input length.

Basic hyperparameter tuning was performed to improve model performance. This includes adjusting parameters such as learning rate, number of iterations, and model complexity. The training process was designed to balance accuracy and computational efficiency.

3.10 Model Implementation

The model development stage focuses on implementing and comparing multiple classification approaches for detecting fake news in environmental and disaster-related datasets. Rather than relying on a single method, this study adopts a comparative modelling framework to evaluate how different algorithms perform in terms of predictive accuracy, interpretability, and computational efficiency.

The selected models represent three major categories of machine learning approaches:

- Traditional machine learning
- Deep learning
- Transformer-based models

This combination allows the study to analyse both simple and advanced techniques within a unified pipeline, providing a balanced evaluation of their strengths and limitations in domain-specific fake news detection.

3.10.1 Logistic Regression

Logistic Regression is implemented as a baseline classification model due to its simplicity, efficiency, and interpretability. It is particularly well suited for text classification tasks when combined with TF-IDF feature representation.

Despite its simplicity, Logistic Regression often performs competitively in text classification tasks, making it a strong reference point for evaluating more complex models.

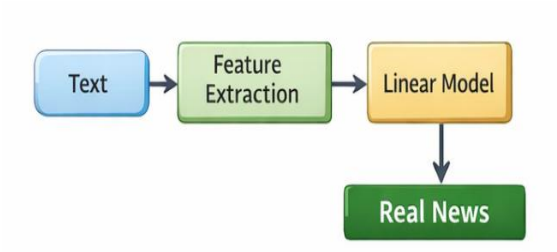

Figure 3.6:Logistic Regression

Figure 3.6 illustrates the workflow of the Logistic Regression model, where textual data is first transformed into numerical features using TF-IDF, followed by a linear classification layer that outputs the predicted class label. The code shown below was used to implement the Logistic Regression model for this study.

```python
lr_pipeline = Pipeline([
    ('tfidf', TfidfVectorizer(max_features=10000, ngram_range=(1,2))),
    ('clf', LogisticRegression(max_iter=1000))
])

start_time = time.time()
lr_pipeline.fit(X_train, y_train)
lr_train_time = time.time() - start_time

y_pred_lr = lr_pipeline.predict(X_test)
y_prob_lr = lr_pipeline.predict_proba(X_test)[:, 1]

print("Logistic Regression")
print("Accuracy:", accuracy_score(y_test, y_pred_lr))
print(classification_report(y_test, y_pred_lr, target_names=['Fake', 'Real']))

plot_confusion_matrix(y_test, y_pred_lr, "Logistic Regression Confusion Matrix")

lr_result = evaluate_model(
    "Logistic Regression",
    y_test,
    y_pred_lr,
    y_prob=y_prob_lr,
```

Figure 3.6.1: Code Implementation of logistic regression model

3.10.2 XGBoost

XGBoost (Extreme Gradient Boosting) is an advanced ensemble learning algorithm that builds a sequence of decision trees to improve prediction accuracy. Each tree is trained to correct the errors of previous models, allowing the algorithm to capture more complex patterns in the data.

In this study, XGBoost is applied to engineered features, including TF-IDF representations and credibility-based features

The model also includes regularization mechanisms that help prevent overfitting, making it suitable for handling high-dimensional text data.

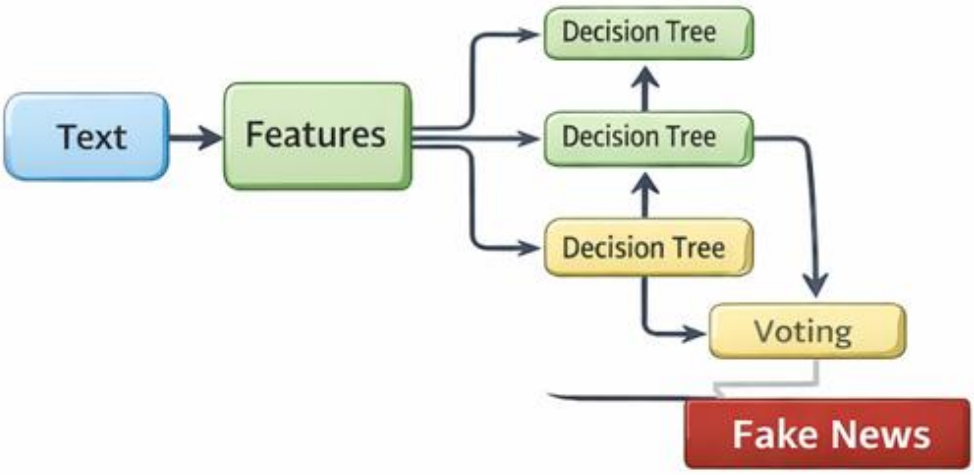

Figure 3.7: Extreme Gradient Boosting (Decision Tree)

Figure 3.7 presents the architecture of the XGBoost model, showing how multiple decision trees are sequentially combined to refine predictions and produce the final classification output.

The code presented below corresponds to the implementation of the XGBoost model employed for classification in this work.

```
xgb_pipeline = Pipeline([
    ('tfidf', TfidfVectorizer(max_features=10000, ngram_range=(1,2))),
    ('clf', XGBClassifier(
        n_estimators=200,
        max_depth=6,
        learning_rate=0.1,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42,
        eval_metric='logloss'
    ))
])

start_time = time.time()
xgb_pipeline.fit(X_train, y_train)
xgb_train_time = time.time() - start_time

y_pred_xgb = xgb_pipeline.predict(X_test)
y_prob_xgb = xgb_pipeline.predict_proba(X_test)[:, 1]

print("XGBoost")
print("Accuracy:", accuracy_score(y_test, y_pred_xgb))
print(classification_report(y_test, y_pred_xgb, target_names=['Fake', 'Real']))

plot_confusion_matrix(y_test, y_pred_xgb, "XGBoost Confusion Matrix")

xgb_result = evaluate_model(
    "XGBoost",
    y_test,
    y_pred_xgb,
    y_prob=y_prob_xgb,
    train_time=xgb_train_time
)
```

Figure 3.7.1: Code Implementation of XGBoost model

3.10.3 BiLSTM

The Bidirectional Long Short-Term Memory (BiLSTM) model is a deep learning approach designed to process text as a sequence of tokens. Unlike traditional models that treat text as independent features, BiLSTM captures the sequential and contextual relationships between words.

By processing text in both forward and backward directions, the model gains a more comprehensive understanding of sentence structure and meaning. This is particularly important in fake news detection, where the interpretation of a sentence may depend on context rather than individual words.

BiLSTM is well suited for identifying linguistic patterns such as tone, phrasing, and narrative flow, which may help distinguish between real and fake news articles.

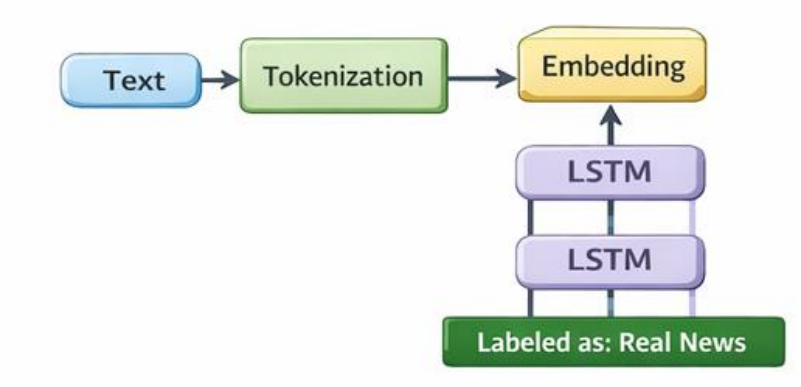

Figure 3.8: Bidirectional Long Short-Term Memory

Figure 3.8 demonstrates the BiLSTM architecture, where input text is tokenized and passed through embedding layers, followed by bidirectional LSTM layers that capture contextual dependencies before classification. The following code illustrates the implementation of the Bidirectional Long Short-Term Memory model used in this study.

```
dl_model = Sequential([
    Embedding(input_dim=max_words, output_dim=128, input_length=max_len),
    Bidirectional(LSTM(64)),
    Dropout(0.5),
    Dense(32, activation='relu'),
    Dropout(0.3),
    Dense(1, activation='sigmoid')
])

dl_model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

dl_model.summary()
```

Figure 3.9.1: Code Implementation of Bidirectional Long Short-Term Memory

3.10.4 DistilRoBERTa

DistilRoBERTa is a transformer-based language model that leverages self-attention mechanisms to understand contextual relationships across entire text sequences. It is a compressed version of the RoBERTa model, designed to provide high performance while reducing computational requirements.

Unlike traditional and recurrent models, DistilRoBERTa captures deep semantic meaning by considering the relationships between all words in a sentence simultaneously. This makes it particularly effective in handling complex and nuanced language, which is common in misleading or deceptive news content.

The model is fine-tuned on the dataset for classification tasks, allowing it to adapt it pretrained language knowledge to the specific domain of environmental and disaster-related news.

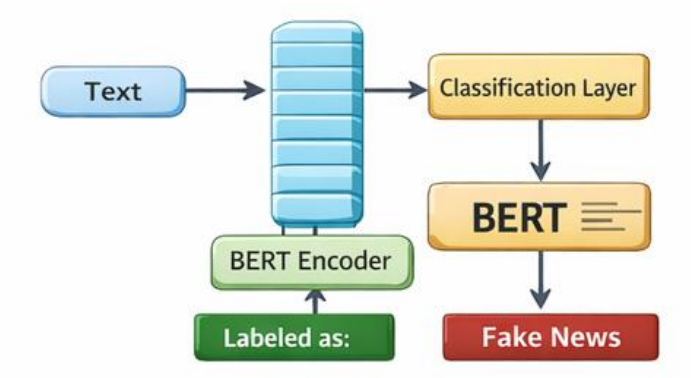

Figure 3.10: DistilRoBERTa Model

Figure 3.9 illustrates the DistilRoBERTa architecture, highlighting the transformer encoder and classification layer used to generate predictions based on contextual embeddings.

The code presented below corresponds to the implementation of the DistilRoBERTa model employed for classification in this work.

```
transformer_eval = trainer.predict(test_dataset)

transformer_logits = transformer_eval.predictions
y_pred_transformer = np.argmax(transformer_logits, axis=1)
y_prob_transformer = torch.nn.functional.softmax(
    torch.tensor(transformer_logits), dim=1
)[:, 1].numpy()

print("DistilRoBERTa Results")
print(classification_report(test_labels, y_pred_transformer, target_names=["Fake", "Real"]))

plot_confusion_matrix(test_labels, y_pred_transformer, "DistilRoBERTa Confusion Matrix")

transformer_result = {
    'Model': 'DistilRoBERTa',
    'Accuracy': accuracy_score(test_labels, y_pred_transformer),
    'Precision': precision_score(test_labels, y_pred_transformer),
    'Recall': recall_score(test_labels, y_pred_transformer),
    'F1-score': f1_score(test_labels, y_pred_transformer),
    'MCC': matthews_corrcoef(test_labels, y_pred_transformer),
    'ROC-AUC': roc_auc_score(test_labels, y_prob_transformer),
    'PR-AUC': auc(
        precision_recall_curve(test_labels, y_prob_transformer)[1],
        precision_recall_curve(test_labels, y_prob_transformer)[0]
    ),
    'Log Loss': log_loss(test_labels, y_prob_transformer),
    'Train Time (s)': transformer_train_time
}
```

Figure 3.11.1: Code Implementation of DistilRoBERTa Model

3.11 Implementation Challenges

During the development of the proposed fake news detection system, several implementation challenges were encountered. These challenges relate to data quality, labelling, model complexity, and computational constraints.

One of the primary challenges was the lack of manually labelled data. Since the dataset did not contain verified labels, a weak labelling approach had to be implemented. Although this method enabled supervised learning, it introduced potential noise and uncertainty in the labels, which may affect model performance.

Another challenge involved data quality and preprocessing. The dataset contained missing values, duplicate records, and inconsistencies across textual and metadata fields. Significant effort was required to clean and standardise the data to ensure it was suitable for model training.

The implementation also faced challenges related to feature representation. Traditional machine learning models required TF-IDF representations, while deep learning models required tokenization and sequence padding. Managing multiple feature formats for different models increased the complexity of the pipeline.

In addition, model training and computational resources posed limitations, particularly for deep learning and transformer-based models such as BiLSTM and DistilRoBERTa. These models require more processing time and memory compared to traditional machine learning approaches, making experimentation and tuning more demanding.

Another challenge was model comparison and evaluation. Ensuring fair comparison across different models required consistent preprocessing, data splitting, and evaluation metrics. Care was taken to maintain uniform evaluation conditions to produce reliable results.

3.12 Summary

This chapter presented the design and implementation of the fake news detection system. It described the key stages of the framework, including data preprocessing, weak labelling, feature engineering, and model development.

The chapter also explained the training setup, evaluation methods, interpretability considerations, and tools used in the implementation. In addition, it outlined the overall system workflow and discussed the challenges encountered during development.

Overall, this chapter provides a foundation for the experimental results presented in the next chapter.

4 Results and Evaluation

This chapter presents the results of the fake news detection models developed in this study and evaluates their performance using multiple metrics. The aim is to assess how effectively different modelling approaches can classify environmental and disaster-related news as real or fake.

4.1 Introduction

The evaluation is conducted using multiple models, including traditional machine learning approaches (Logistic Regression and XGBoost), deep learning models (BiLSTM), and transformer-based models (DistilRoBERTa). A baseline model is also included to provide a reference for performance comparison.

The chapter begins by describing the experimental setup, including dataset splitting and evaluation metrics. This is followed by an exploratory data analysis to understand the characteristics of the dataset. Subsequently, the performance of each model is presented and compared using standard classification metrics. Additional analysis, including confusion matrices and performance curves, is used to provide deeper insights into model behaviour.

Finally, the results are discussed in relation to the research objectives, highlighting key findings, strengths, and limitations of the proposed approach.

4.1.1 Dataset Split

The dataset is split into training (80%) and testing (20%) sets using a standard train-test approach. The same split and preprocessing steps are applied across all models, including Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa, ensuring fair comparison. Figure 4.1 illustrates this process.

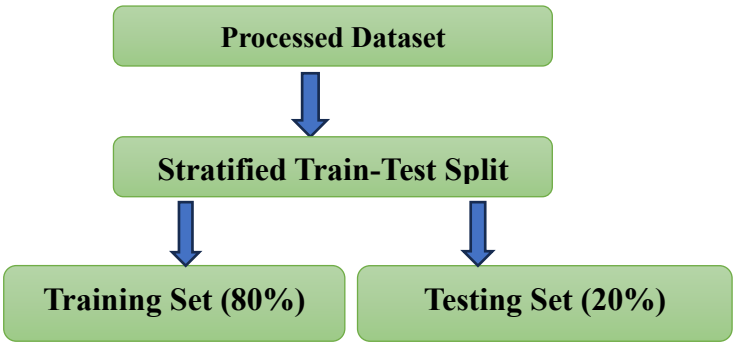

Figure 4.1: Dataset Splitting into Training and Testing Sets

Figure 4.1 illustrates the dataset splitting process used in this study. After preprocessing and weak labelling, the final dataset is divided into training and testing sets using an 80:20 split. A stratified split is applied to preserve the class distribution in both subsets, ensuring that the evaluation remains

fair and representative. The training set is used for model learning, while the testing set is reserved for performance assessment on unseen data.

4.1.2 Evaluation Metrics

The performance of the models is evaluated using a range of standard classification metrics to provide a comprehensive assessment.

Accuracy is used to measure the overall proportion of correctly classified instances. However, since accuracy alone may not fully capture model performance, additional metrics are considered.

Precision measures the proportion of correctly predicted positive instances, indicating how reliable the model is when predicting fake news. Recall measures the proportion of actual fake news instances that are correctly identified, reflecting the model's ability to detect misleading content. The F1-score combines precision and recall into a single metric, providing a balanced measure of performance.

In addition to these metrics, ROC-AUC and PR-AUC are used to evaluate the model's ability to distinguish between classes across different threshold values. These metrics provide deeper insight into model performance, particularly in classification tasks where class imbalance may be present.

Together, these evaluation metrics ensure a comprehensive and reliable comparison of the different models used in this study.

Metric	**Description**
Accuracy	Overall correctness
Precision	Correct fake predictions
Recall	Ability to detect fake news
F1-score	Balance of precision and recall
ROC-AUC	Classification ability
PR-AUC	Performance on positive class

Table 4.1:Evaluation Metrics

4.2 Exploratory Data Analysis

An exploratory data analysis (EDA) was conducted to understand the structure, distribution, and key characteristics of the dataset prior to model training. This step is essential as it provides insights into class distribution, labelling behaviour, and potential data imbalances that may influence model performance. The dataset consists of environmental and disaster-related news articles labelled using a weak labelling approach, with three categories: *Real*, *Fake*, and *Unknown*.

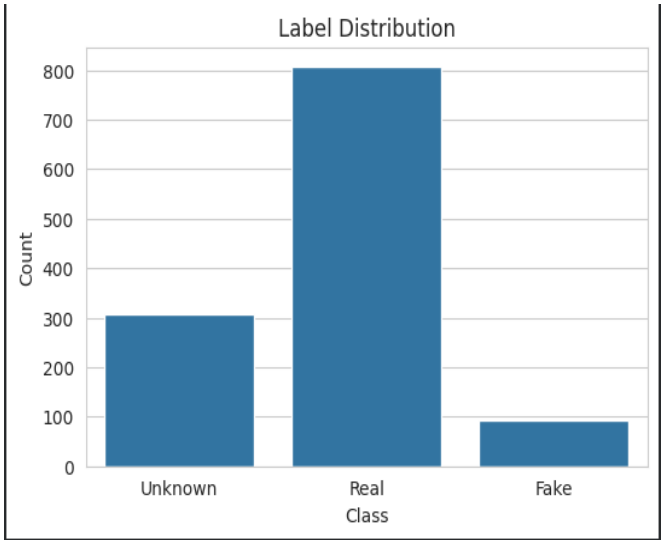

Figure 4.2: Distribution of Weakly Labelled News Articles

As shown in Figure 4.1, the dataset is not evenly distributed across the three classes. The majority of articles are labelled as *Real*, followed by a smaller proportion labelled as *Unknown*, and a significantly smaller number labelled as *Fake*.

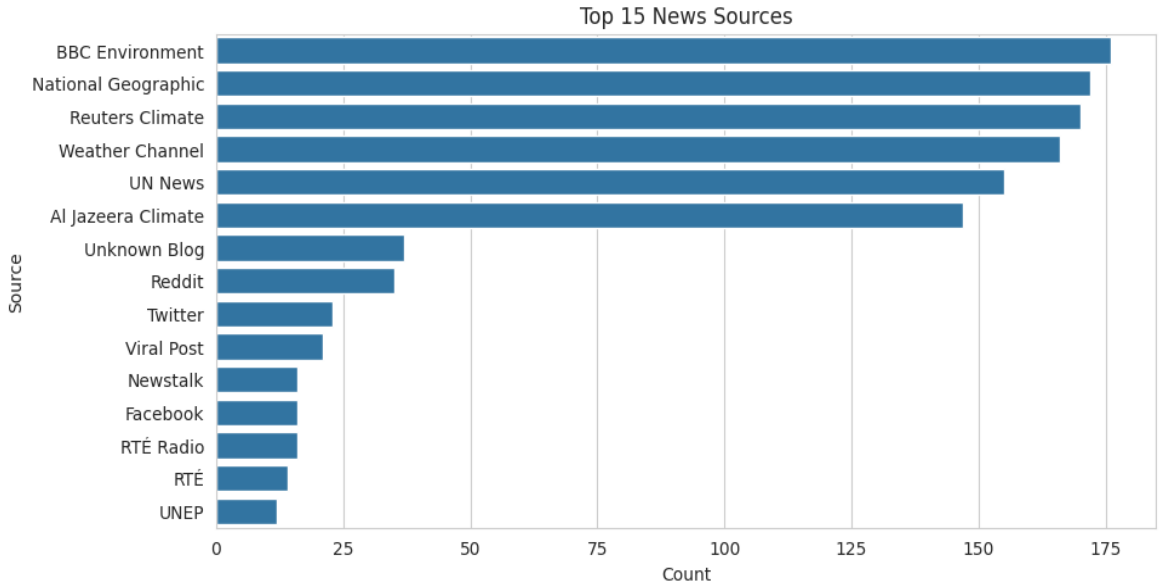

Figure 4.2: News Sources in the Dataset

Figure 4.2 presents the most frequent news sources in the dataset. This analysis provides important context, as source information plays a central role in the weak labelling strategy. A concentration

of articles from a limited number of sources may influence label distribution and affect the contribution of source-based credibility features during model training.

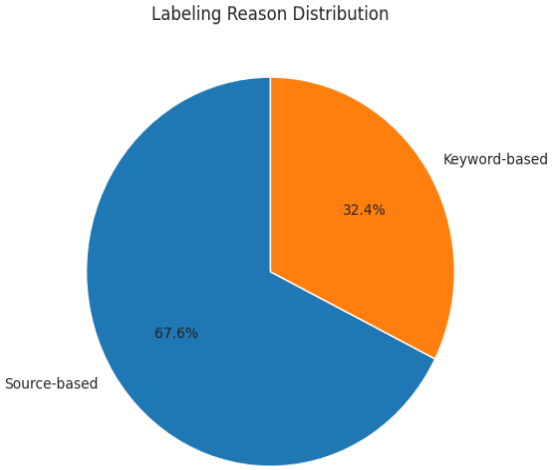

Figure 4.3: Distribution of Labelling Strategies

Figure 4.3 illustrates the distribution of labelling strategies used in the dataset. The results show that source-based labelling accounts for the majority of assigned labels, while keyword-based rules provide supplementary support. This indicates that source credibility is the dominant factor in the weak labelling process, with textual heuristics contributing to the identification of misleading content.

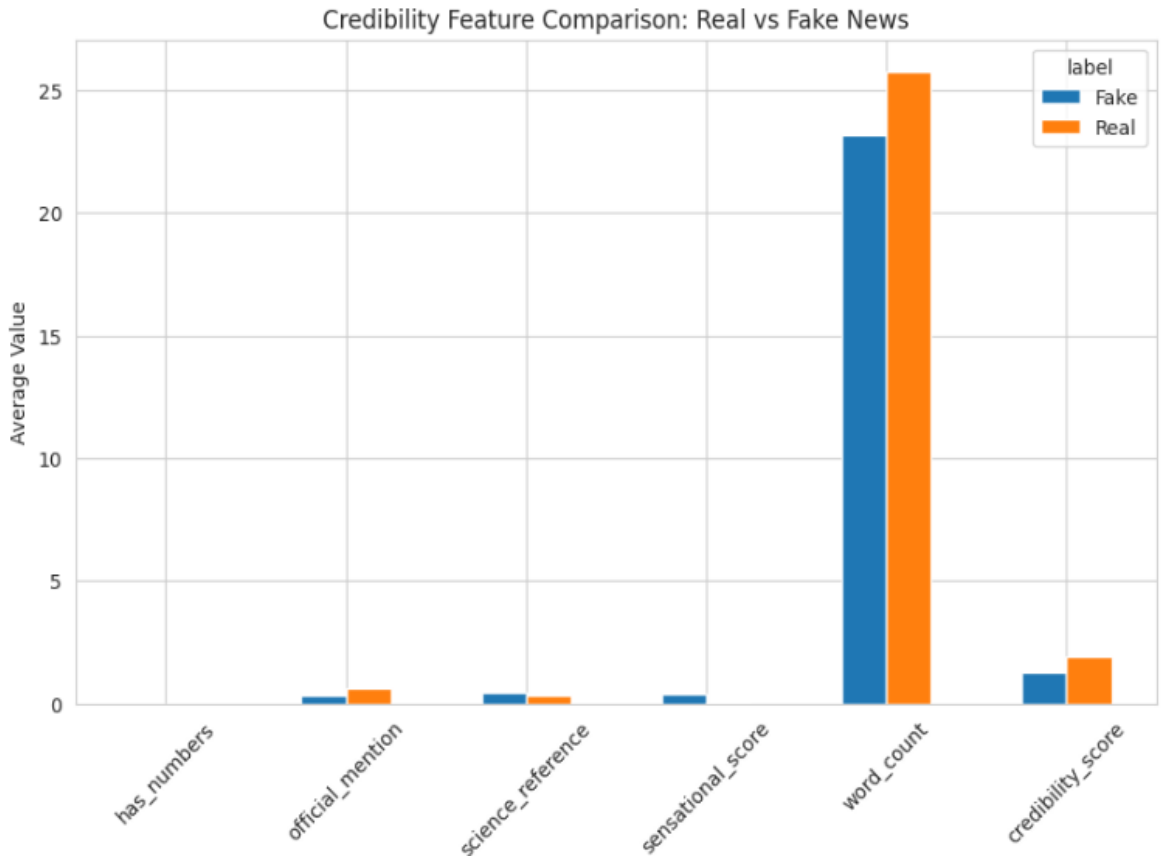

Figure 4.4: Credibility Feature Comparison Between Real and Fake News

Figure 4.4 presents a comparison of credibility-related features between real and fake news articles. Features such as the presence of numerical data, references to official entities, scientific terminology, and sensational language reveal distinct patterns across classes. These differences support the classification process and directly relate to the study objective of identifying features that effectively distinguish real news from fake news.

Overall, the exploratory analysis confirms that the dataset exhibits class imbalance and relies heavily on source-based labelling, both of which are important considerations for model performance and evaluation.

4.3 Model Performance Results

This section presents the quantitative performance of the implemented models in classifying environmental and disaster-related news articles as real or fake. The models evaluated include Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa. Each model was trained using the same dataset split and evaluated on the same test set to ensure consistency and fairness.

The performance metrics used include accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, Matthews Correlation Coefficient (MCC), and log loss. These metrics provide a comprehensive view of model effectiveness, particularly in the presence of class imbalance.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.88	0.91	0.96	0.93	0.78
XGBoost	0.87	0.93	0.93	0.93	0.75
BiLSTM	0.87	0.91	0.95	0.93	0.78
DistilRoBERTa	0.90	0.90	1.00	0.94	0.79

Table 4.2: Model Performance Comparison

Table 4.2 presents a comparative evaluation of four models Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa using multiple performance metrics.

Logistic Regression achieved an accuracy of 0.88, with a precision of 0.91, recall of 0.96, and an F1-score of 0.93. These results indicate a well-balanced performance, with particularly strong recall

suggesting that the model is effective at correctly identifying relevant instances. The ROC-AUC score of 0.78 reflects a moderate ability to distinguish between classes.

XGBoost recorded a slightly lower accuracy of 0.87, but achieved the highest precision (0.93) among all models, indicating a strong ability to minimize false positives. Its recall (0.93) and F1-score (0.93) demonstrate consistent and balanced classification performance. However, its ROC-AUC score of 0.75 is comparatively lower, suggesting reduced effectiveness in overall class separation.

The BiLSTM model also achieved an accuracy of 0.87, with a precision of 0.91 and a recall of 0.95, resulting in an F1-score of 0.93. The relatively high recall indicates improved capability in capturing relevant instances, likely due to its ability to model sequential and contextual dependencies within text. Its ROC-AUC score of 0.78 is comparable to that of Logistic Regression.

DistilRoBERTa demonstrated the best overall performance, achieving the highest accuracy of 0.90. It attained a recall of 1.00, indicating perfect identification of relevant instances, while maintaining a strong precision of 0.90. This resulted in the highest F1-score of 0.94 among all models. Additionally, it achieved the highest ROC-AUC score of 0.79, indicating superior discriminative capability.
The results show a clear improvement across model types. Logistic Regression establishes a strong baseline using TF-IDF features, while XGBoost enhances precision by capturing more complex patterns. The BiLSTM model improves recall through better contextual understanding. DistilRoBERTa achieves the best overall performance, demonstrating the advantage of transformer-based models in capturing deep semantic and contextual information for fake news detection.

4.4 Comparative Evaluation of Models

This section compares the performance of Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa using the metrics in Table 4.2, supported by visual analysis from Python-generated performance charts.

As shown in Figure 4.5, which illustrates the accuracy comparison across all models, there is a minimal variation in overall classification performance. Logistic Regression provides a strong baseline, demonstrating stable performance using TF-IDF features. XGBoost improves upon this by capturing more complex, non-linear relationships between features.

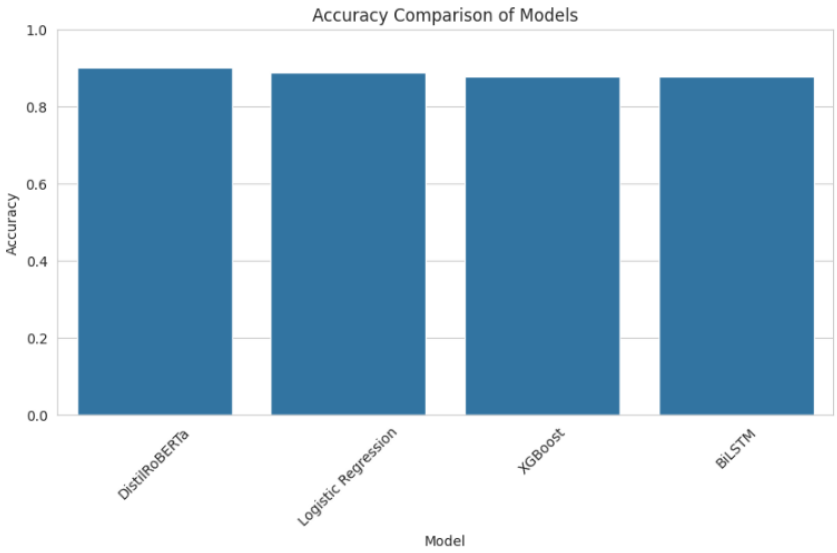

Figure 4.5: Accuracy Comparison of Models

The BiLSTM model further enhances performance by learning sequential dependencies in the text. DistilRoBERTa achieves the highest accuracy among all models, highlighting the effectiveness of transformer-based contextual representations.

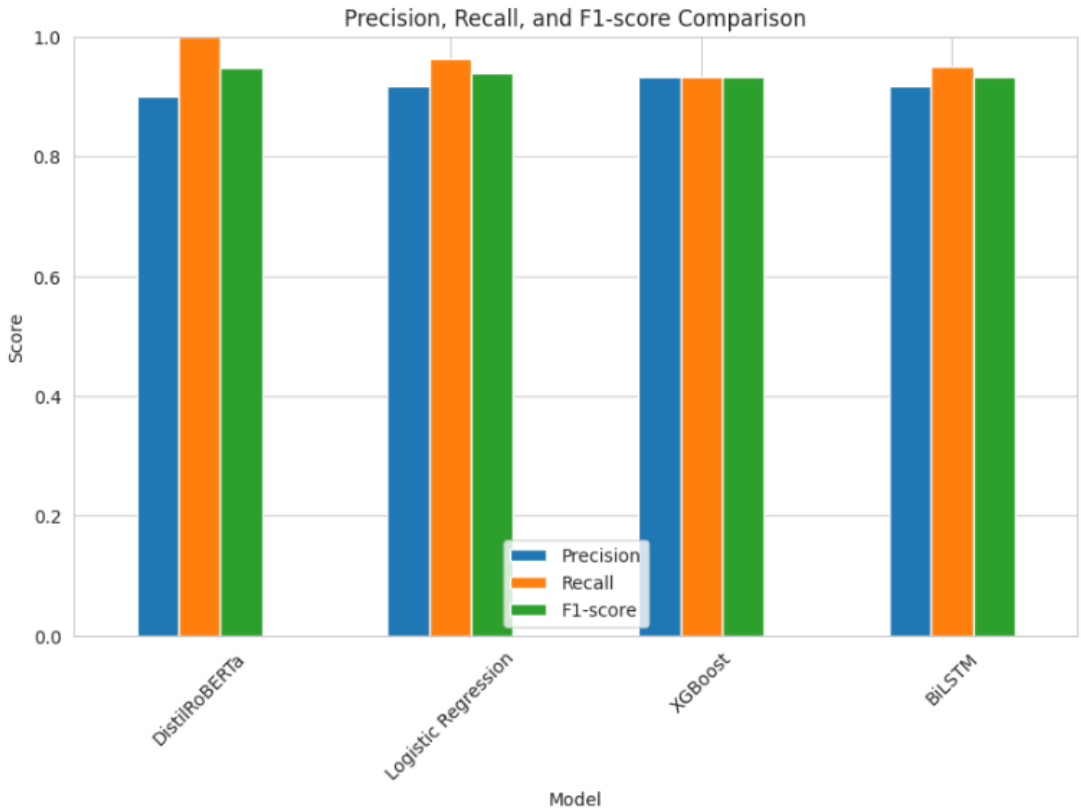

Figure 4.6: Precision, Recall, and F1-score Comparison Across Models

A detailed comparison in Figure 4.6 presents precision, recall, and F1-score for each model. Logistic Regression and XGBoost show moderate performance, while BiLSTM improves recall and F1-score, indicating better fake news detection. DistilRoBERTa outperforms all models across these metrics, achieving the highest classification accuracy.

Further evaluation is provided in Figure 4.7, which presents ROC-AUC, PR-AUC, and Matthews Correlation Coefficient (MCC) for each model. These metrics offer deeper insight into classification performance, especially in the presence of class imbalance.

The results show that DistilRoBERTa consistently achieves the highest scores across all advanced metrics, followed by BiLSTM. Traditional models, while effective, perform comparatively lower in capturing complex patterns in the data.

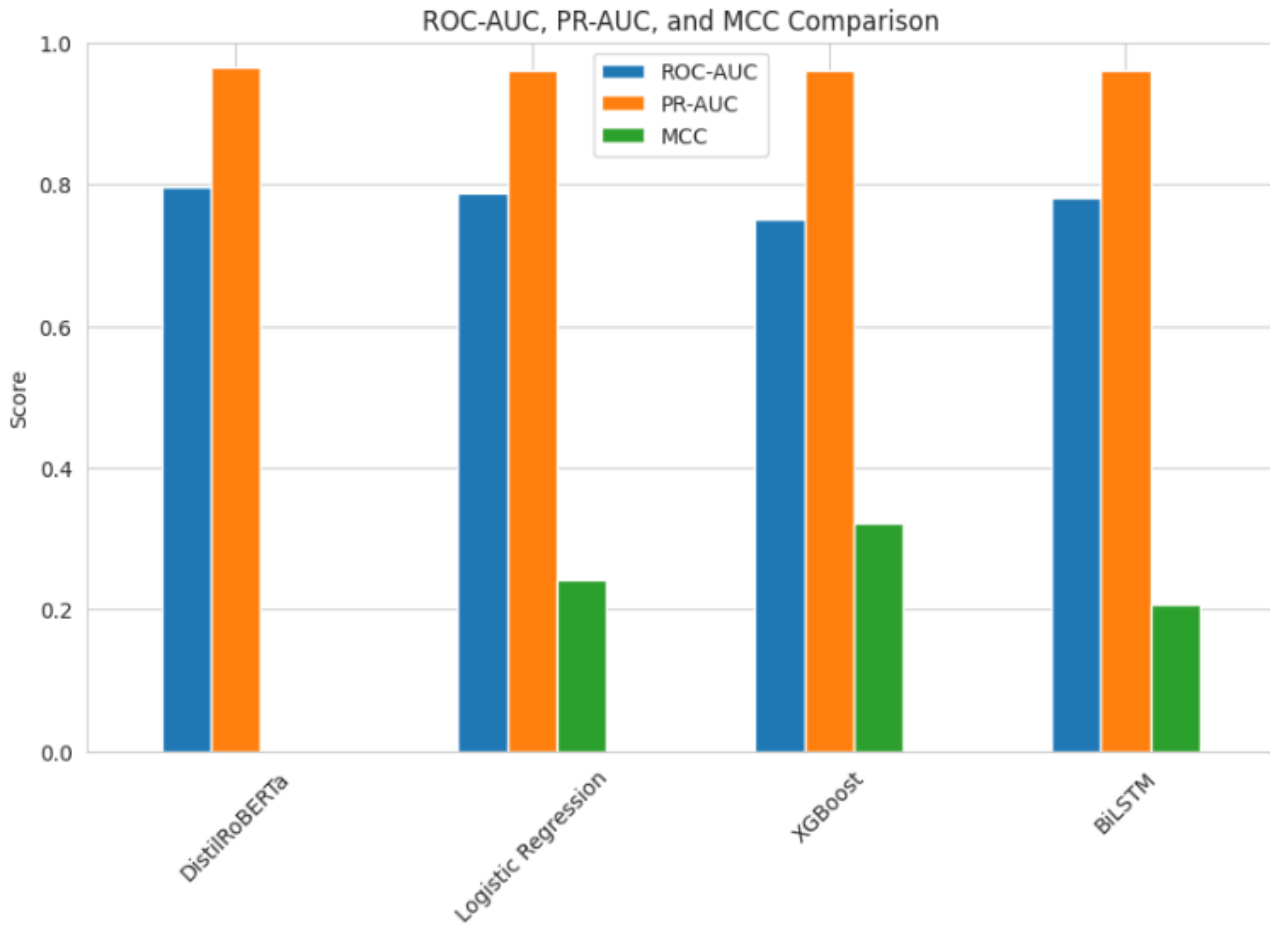

Figure 4.7:ROC-AUC, PR-AUC, and MCC Comparison Across Models

Overall, the comparative analysis demonstrates a minor progression in performance from traditional machine learning models to deep learning and transformer-based approaches. Logistic Regression and XGBoost remain valuable due to their interpretability and computational efficiency, while BiLSTM provides improved contextual learning. However, DistilRoBERTa emerges as the most effective model due to its superior ability to capture semantic and contextual relationships within the text.

These findings confirm that transformer-based models are better suited for domain-specific fake news detection tasks, particularly when dealing with complex and nuanced textual data.

4.5 Confusion Matrix and Curve Analysis

This section analyses model performance using confusion matrices and performance curves, providing deeper insight beyond overall metrics like accuracy and F1-score.

Confusion matrices show true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). In fake news detection, false negatives are critical as they represent undetected misleading content.

Figure 4.8 presents the confusion matrices for all models. Logistic Regression and XGBoost show moderate misclassification, particularly higher false negatives. BiLSTM improves performance with more correct detections and fewer errors.

DistilRoBERTa achieves the most accurate classification results, as shown in Figure 4.8, with the highest concentration of values along the diagonal of the confusion matrix. This indicates that the model produces the largest number of correct predictions and the lowest number of classification errors. In particular, DistilRoBERTa significantly reduces false negatives, making it more reliable for detecting fake news compared to the other models.

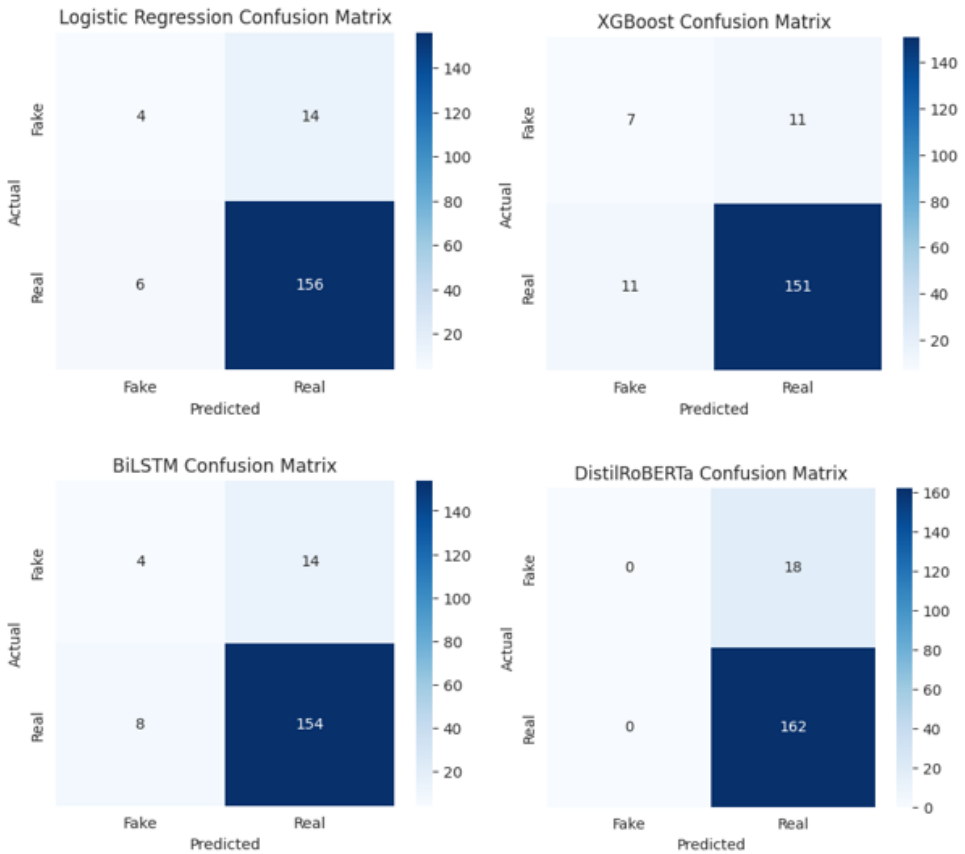

Figure 4.8:Confusion Matrices across Models

In addition to confusion matrices, performance curves were analysed to evaluate model effectiveness across different classification thresholds. The Receiver Operating Characteristic (ROC) curve is used to assess the model's ability to distinguish between real and fake classes. A higher Area Under the Curve (ROC-AUC) indicates better classification performance.

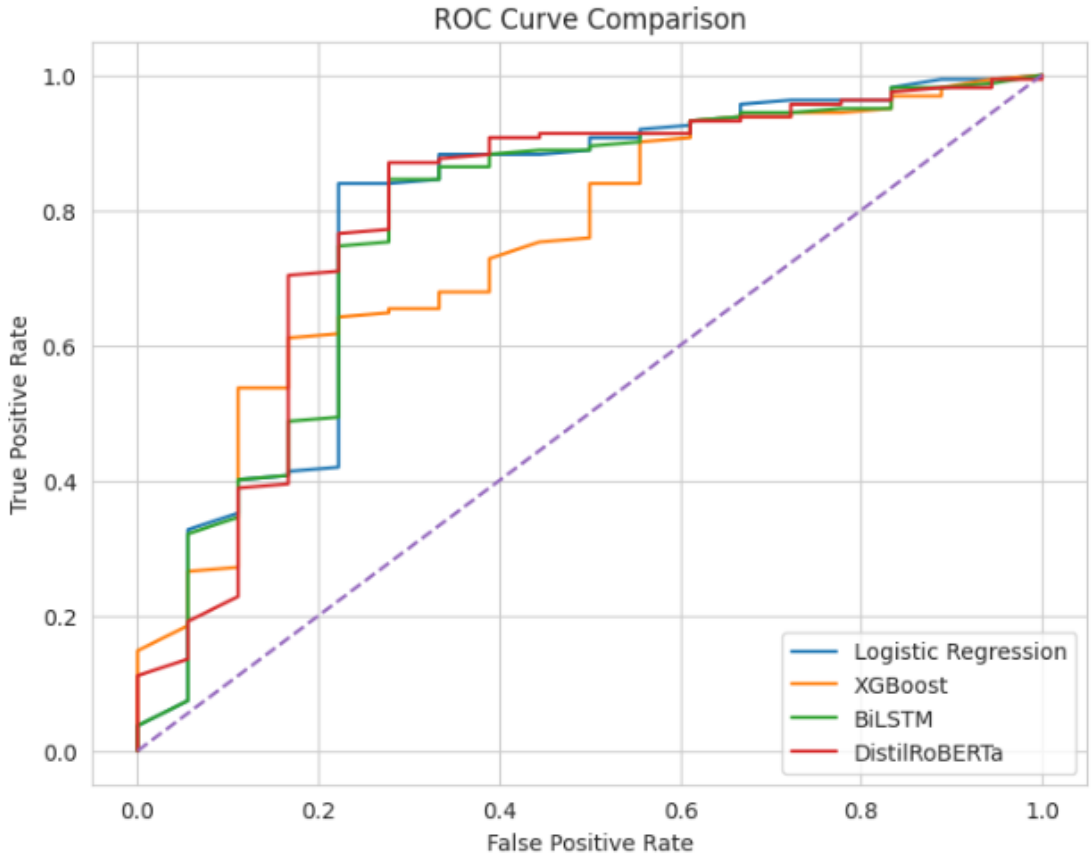

Figure 4.9:ROC Curve Comparison across Models

As illustrated in Figure 4.9, all models perform better than random classification; however, DistilRoBERTa achieves the highest ROC-AUC score, followed by BiLSTM. Logistic Regression and XGBoost show comparatively lower performance, reflecting their limitations in capturing complex patterns in the data.

The Precision-Recall (PR) curve, shown in Figure 4.10, provides additional insight into model performance, particularly in handling class imbalance. The PR curve focuses on the trade-off between precision and recall for the positive class (fake news). The results indicate that DistilRoBERTa maintains higher precision across a range of recall values, demonstrating its effectiveness in correctly identifying fake news while minimising false predictions.

Overall, the confusion matrix and curve analysis confirm that transformer-based models outperform traditional and deep learning approaches in both accuracy and reliability. DistilRoBERTa not only

achieves higher overall performance but also reduces critical errors such as false negatives, making it the most suitable model for fake news detection in this study.

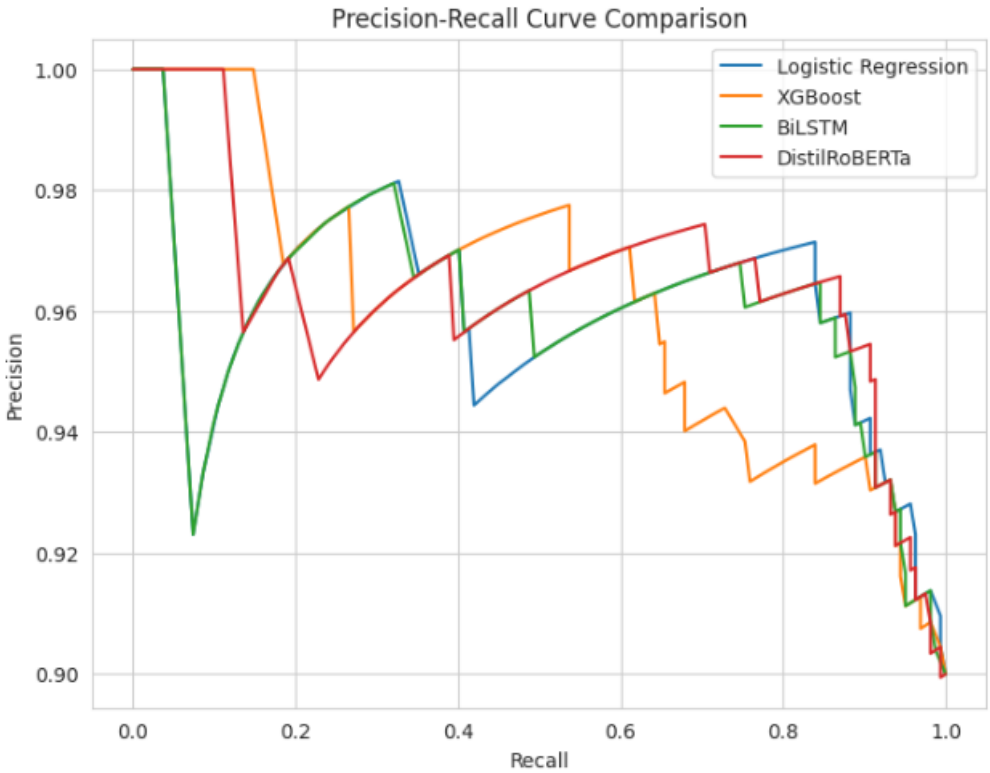

Figure 4.10: Precision-Recall Curve Comparison across Models

4.6 Limitations of the Results

While the results of this study demonstrate the effectiveness of the proposed fake news detection framework, several limitations must be acknowledged when interpreting the findings.

One of the primary limitations relates to the use of a weakly labelled dataset. The labels in this study were generated using a rule-based approach based on source credibility and keyword patterns, rather than manual verification. Although this method enables the use of supervised learning in the absence of annotated data, it may introduce noise and inaccuracies in the labels. As a result, the evaluation metrics may not fully reflect true classification performance.

The study is also limited by its domain-specific focus. The dataset consists exclusively of environmental and disaster-related news articles, which means that the findings may not generalise to other domains such as political, financial, or health-related misinformation. Language patterns, credibility indicators, and content characteristics can vary significantly across domains, which may affect model performance when applied to different types of data.

In addition, the models were trained and evaluated on English-language text only. This restricts the applicability of the results to multilingual contexts, where linguistic structures and expressions differ. Extending the framework to support multiple languages would require additional preprocessing and model adaptation.

Another limitation is related to computational requirements. Advanced models such as BiLSTM and DistilRoBERTa require significantly more computational resources and training time compared to traditional machine learning models. This may limit their practicality in real-time or resource-constrained environments.

Finally, the study focuses on a research-based implementation rather than a fully deployed system. The models were evaluated in a controlled experimental setting, and real-world deployment considerations such as scalability, real-time processing, and integration with external systems were not addressed.

Despite these limitations, the results provide valuable insights into the application of machine learning and transformer-based models for fake news detection in a domain-specific context. These limitations also highlight areas for further research and improvement, which are discussed in the following chapter.

5 Conclusion and Future Work

5.1 Introduction

This chapter presents the conclusion of the study and reflects on the overall outcomes of the research. It summarises the key contributions, discusses how the research objectives have been achieved, and evaluates the effectiveness of the proposed approach. In addition, the chapter highlights the limitations of the study and provides recommendations for future work in the area of fake news detection.

5.2 Summary of the Research

This study aimed to develop and evaluate a domain-specific fake news detection framework for environmental and disaster-related news using weak supervision and multiple modelling approaches. The motivation stems from the growing impact of misinformation in critical areas where inaccurate information can have serious consequences.

A structured dataset was prepared, including textual content and metadata such as source, author, and date. Due to the lack of manual labels, a weak labelling strategy was applied, combining source credibility and domain-specific keyword analysis to generate usable classifications under realistic constraints.

The study implemented a comprehensive processing pipeline that included data cleaning, preprocessing, feature engineering, and model development. Textual features were represented using TF-IDF for traditional machine learning models, while token-based representations were used for deep learning and transformer-based approaches. In addition, customised credibility features were introduced to capture domain-relevant indicators of reliability, enhancing the model's ability to distinguish between real and fake news.

A comparative framework was used, including Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa, representing different levels of complexity and learning approaches. All models were evaluated under a consistent setup using multiple metrics to ensure a fair and comprehensive comparison.

Overall, the study shows that a structured, domain-specific approach, combining weak labelling and multi-model evaluation, can effectively detect fake news even without fully annotated data. It also highlights the importance of credibility-based features and comparative analysis in improving model performance.

5.3 Key Findings

The study produced several important findings related to fake news detection in environmental and disaster-related news.

First, the use of weak labelling proved to be a practical and effective approach for generating training data in scenarios where manual annotation is not feasible. Although the labels may contain some noise, the combination of source credibility and keyword-based signals provided a reasonable approximation of ground truth, enabling supervised learning models to perform effectively.

Second, the comparative evaluation of models showed clear differences in performance and behaviour. Traditional machine learning models such as Logistic Regression demonstrated strong baseline performance, particularly when combined with TF-IDF features. XGBoost provided improved performance by modelling non-linear relationships and interactions between features, particularly when credibility-based features were included. This highlights the importance of feature engineering in enhancing model effectiveness.

Deep learning and transformer-based models, particularly DistilRoBERTa, showed strong capability in capturing contextual and semantic relationships within text. These models were more effective in handling complex language patterns and subtle forms of misinformation. However, they required greater computational resources and offered lower interpretability compared to traditional models.

Third, the integration of customised credibility features contributed positively to classification performance. Features such as source reliability, keyword patterns, and credibility scoring provided additional signals beyond textual content, improving the model's ability to distinguish between real and fake news.

Finally, the results highlight that no single model is universally optimal. Instead, there is a trade-off between performance, interpretability, and computational efficiency. This reinforces the importance of comparative evaluation when selecting models for practical applications.

5.4 Achievement of Aim and Objectives

The primary aim of this research-to design, implement, and evaluate a comparative fake news detection framework for environmental and disaster-related news using weak labelling has been successfully achieved.

An end-to-end framework was developed, including preprocessing, weak labelling, feature engineering, model development, and evaluation. The study addressed key challenges such as the

lack of fully annotated datasets through a rule-based labelling approach using source credibility and domain-specific keywords.

All objectives were met, including literature review, dataset preparation, feature engineering, and implementation of multiple models (Logistic Regression, XGBoost, BiLSTM, and DistilRoBERTa). A consistent evaluation framework enabled effective comparison across models.

Results show that the framework can accurately classify news even with weakly labelled data, though performance depends on labelling quality and domain characteristics. Overall, the study provides a practical and adaptable approach for domain-specific fake news detection.

5.5 Answer to the Research Question

The primary research question guiding this study was:

"Can machine learning and transformer-based models classify environmental and disaster-related news as real or fake using a weakly labelled dataset?"

The results of the main Objective of this study demonstrate that machine learning, deep learning, and transformer-based models are capable of effectively classifying environmental and disaster-related news using a weakly labelled dataset. Despite the absence of manually annotated ground truth labels, the models were able to learn meaningful patterns from the data and produce reliable classification outcomes.

Transformer-based models, particularly DistilRoBERTa, showed strong performance due to their ability to capture contextual and semantic relationships within the text. In contrast, traditional machine learning models such as Logistic Regression and XGBoost provided stable baseline results with the added advantage of interpretability and lower computational requirements.

The findings indicate that a rule-based weak labelling approach is both practical and effective for generating classification labels in the absence of manually annotated data. The use of source credibility and keyword-based scoring enabled the systematic assignment of Real, Fake, and Unknown labels.

Although weak labelling inherently introduces some level of noise, the results demonstrate that the generated labels were sufficiently reliable for training classification models. Furthermore, the exclusion of Unknown instances from the training dataset contributed to improved label consistency and model performance.

This confirms that weak labelling in relation to Objective (a) is a viable and scalable approach for handling real-world datasets where manual annotation is limited or unavailable.

The results highlight the importance of both textual and credibility-related features in distinguishing real news from fake news in relation to Objective (b). Textual features, including TF-IDF representations, n-grams, and contextual embeddings, enabled the models to capture linguistic patterns such as vocabulary usage, tone, and writing style. These features were particularly useful in identifying differences between factual reporting and sensational or misleading content.

In addition, credibility-related features, such as source reliability and keyword-based indicators, played a significant role in improving classification performance. Articles originating from verified sources and those containing factual or domain-specific keywords were more likely to be classified as real, whereas articles associated with unverified sources and sensational language were more likely to be classified as fake.

The integration of both feature types enhanced the overall effectiveness of the models, demonstrating the value of combining linguistic and credibility-based signals in fake news detection.

The results show clear performance differences across models. Traditional models like Logistic Regression and XGBoost provide consistent, interpretable results with low computational cost. BiLSTM improves contextual understanding but requires more resources and training time.

Transformer models, especially DistilRoBERTa, achieve the best performance by capturing complex semantic relationships, though at the cost of higher complexity and lower interpretability.

Overall, the findings highlight trade-offs between performance, efficiency, and interpretability. The study confirms that weakly labelled data can be effectively used for fake news detection, and that combining textual and credibility features enhances model performance.

5.6 Limitations of the Study

Despite the positive outcomes of this research, several limitations should be acknowledged, as they influence the interpretation and generalisability of the findings.

A key limitation of this study is its reliance on weak labelling. While this approach enables dataset creation without manual annotation, it introduces potential noise and uncertainty. Labels based on heuristics, such as source credibility and keyword patterns, may not always reflect true content, which can impact classification accuracy.

Another limitation relates to the domain-specific nature of the dataset. While focusing on environmental and disaster-related news provides valuable insights, the findings may not generalise to other domains such as political, health, or social misinformation. Different domains often exhibit distinct linguistic patterns, credibility indicators, and misinformation strategies, which may require tailored approaches.

The study is also limited to English-language data, which restricts its applicability in multilingual contexts. Fake news is a global issue, and models trained solely on English text may not perform effectively when applied to content in other languages or cultural settings.

Another important limitation is related to computational resources. Advanced models such as BiLSTM and DistilRoBERTa require substantial processing power and training time, which may restrict their deployment in resource-constrained environments. This highlights a trade-off between model performance and practical usability.

Finally, while the study incorporates credibility-based features, it does not consider additional contextual signals such as social network interactions, user behaviour, or dissemination patterns. These factors are increasingly recognised as important components in comprehensive fake news detection systems.

5.7 Recommendations for Future Work

This study provides a foundation for fake news detection in environmental and disaster-related domains; however, several areas can be further improved in future research.

First, the labelling process can be enhanced by combining weak labelling with a subset of manually annotated data. This hybrid approach would improve label quality and reduce noise, leading to more reliable model performance.

Second, future work should expand the dataset to include multiple domains and languages. Incorporating multilingual data and diverse topics such as political or health misinformation would improve the generalisability of the framework.

Third, additional features beyond textual and credibility-based signals could be explored. Integrating social media metadata, user engagement patterns, and network-based information may provide deeper insights into misinformation detection.

Another important direction is the optimisation of deep learning and transformer-based models. While models such as DistilRoBERTa offer strong performance, their computational cost may limit

practical use. Future research could focus on more efficient architectures or model compression techniques.

For future work, this methodology can be applied to additional datasets to assess its generalisability and robustness, as well as to analyse its performance across diverse data distributions and domains.

In addition, implementing the system in real-time environments would allow evaluation under dynamic conditions, where misinformation evolves continuously. This would improve understanding of the system's practical effectiveness.

Finally, improving model interpretability is essential. Applying explainable AI techniques would enhance transparency and help users understand how and why classifications are made.

Overall, these directions support the development of more robust, scalable, and practical fake news detection systems.

5.8 Final Conclusion

This research presented a domain-specific fake news detection framework for environmental and disaster-related news, integrating weak labelling, feature engineering, and comparative modelling approaches. The study demonstrated that both traditional machine learning and transformer-based models can effectively classify news articles as real or fake, even in the absence of fully annotated datasets. The findings highlight the importance of combining textual representations with credibility-based features, such as source reliability and domain-specific keyword patterns, to improve classification performance. While advanced models such as DistilRoBERTa achieved stronger contextual understanding, simpler models like Logistic Regression remained valuable due to their efficiency and interpretability.

Overall, the research confirms that a structured and domain-focused approach can provide a practical solution to fake news detection in resource-constrained settings. Although limitations exist, particularly in relation to weak labelling and dataset scope, the study offers a scalable and adaptable framework that can be extended in future work.

In conclusion, this work contributes to the growing field of misinformation detection by demonstrating the effectiveness of combining weak supervision with multi-model evaluation, while emphasising the need for domain-specific analysis in addressing real-world challenges. Although this work may not be perfect, it has been a valuable learning journey that has enabled me to develop both theoretical understanding and practical skills. Through this thesis, I have gained significant knowledge and experience, which will contribute to my personal growth and support my future career development.

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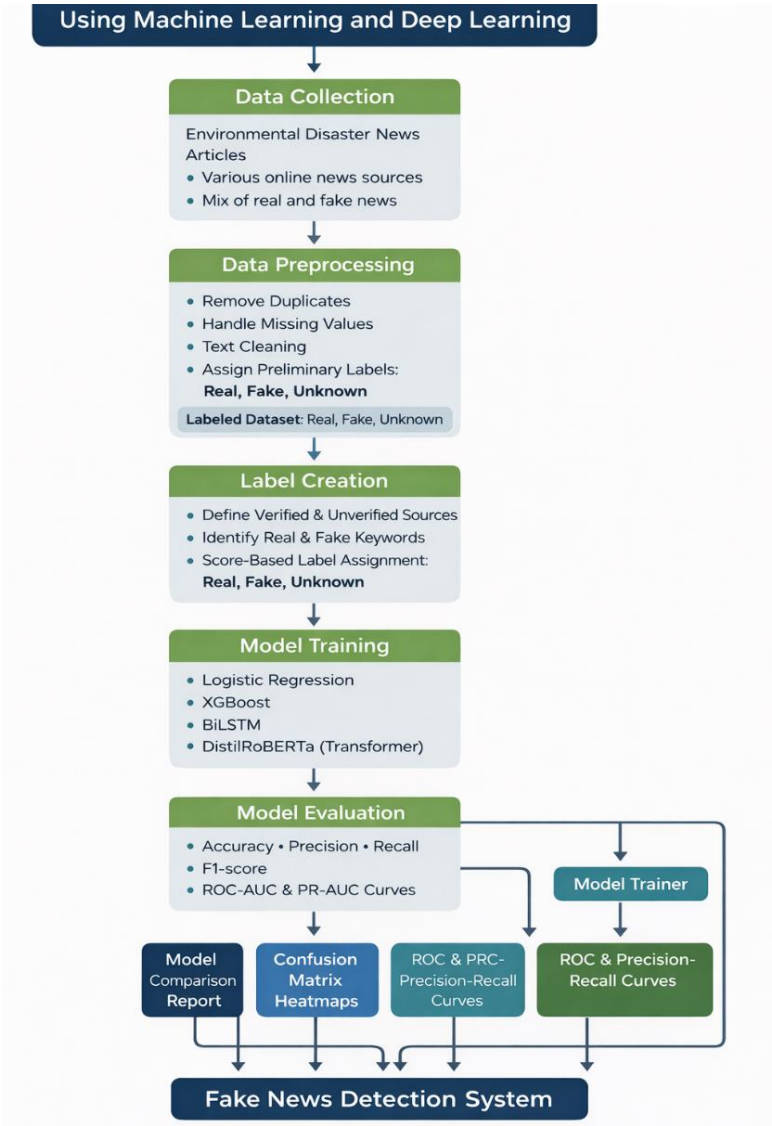
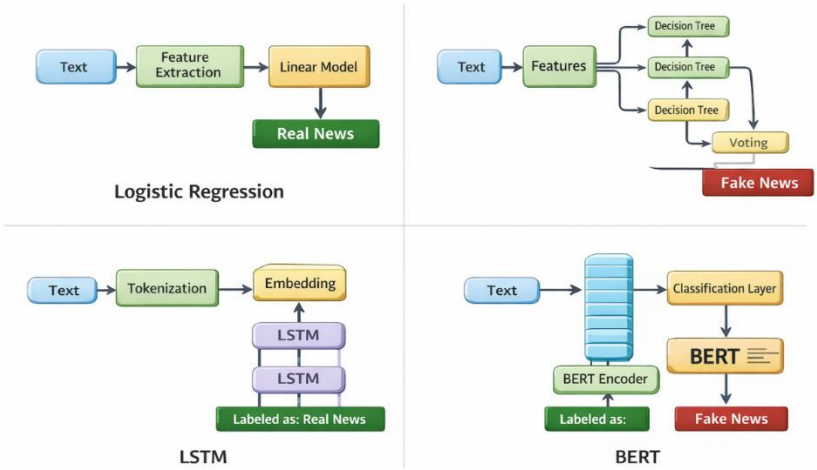
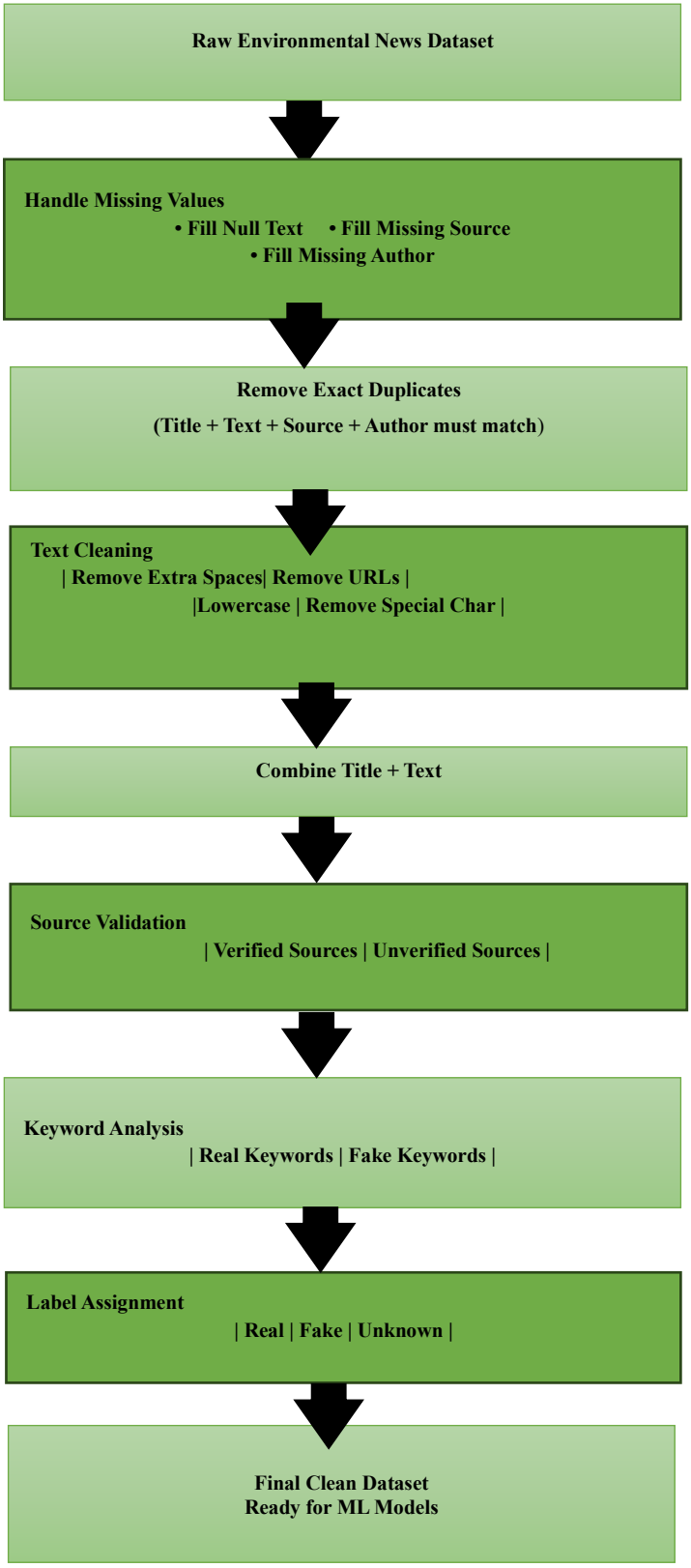
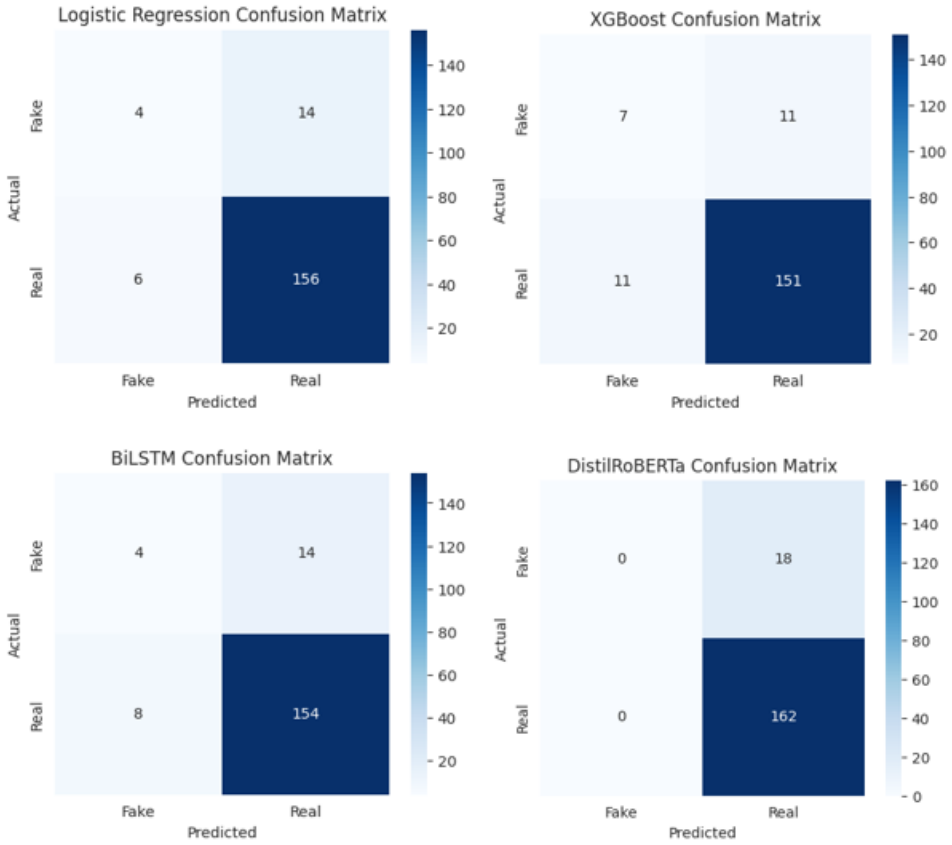
Appendices

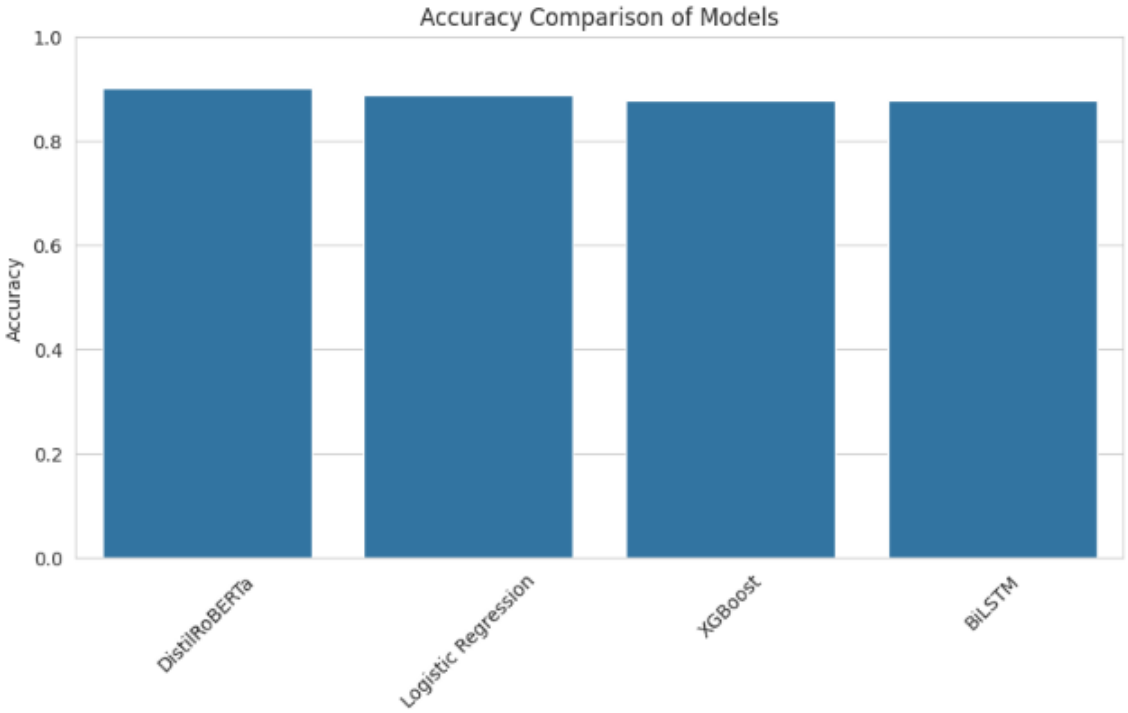
Appendices A: Diagrams






Raw Environmental News Dataset
Handle Missing Values
• Fill Null Text • Fill Missing Source
• Fill Missing Author
Remove Exact Duplicates
(Title + Text + Source + Author must match)
Text Cleaning
| Remove Extra Spaces| Remove URLs |
|Lowercase | Remove Special Char |
Combine Title + Text
Source Validation
| Verified Sources | Unverified Sources |
Keyword Analysis
| Real Keywords | Fake Keywords |
Label Assignment
| Real | Fake | Unknown |
Final Clean Dataset
Ready for ML Models


Logistic Regression Confusion Matrix
Actual
Fake
Real
4
14
6
156
Fake
Real
Predicted
140
120
100
80
60
40
20
XGBoost Confusion Matrix
7
11
11
151
BiLSTM Confusion Matrix
4
14
8
154
DistilRoBERTa Confusion Matrix
0
18
0
162
160
0


Accuracy Comparison of Models
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
DistilRoBERTa
Logistic Regression
XGBoost
BiLSTM

Appendices B: Code

Check the generated labels

```
print(df['label'].value_counts())

print(df[['source', 'label', 'label_reason', 'real_score', 'fake_score']].head(10))

plt.figure(figsize=(6,4))

sns.countplot(x='label', data=df)

plt.title("Label Distribution")

plt.xlabel("Class")

plt.ylabel("Count")

plt.show()

top_sources = df['source'].value_counts().head(15)

plt.figure(figsize=(10,5))

sns.barplot(x=top_sources.values, y=top_sources.index)

plt.title("Top 15 News Sources")

plt.xlabel("Count")

plt.ylabel("Source")

plt.show()

source_based = df['label_reason'].str.contains('source', case=False, na=False).sum()

keyword_based = df['label_reason'].str.contains('keywords', case=False, na=False).sum()


reason_counts = pd.Series({

    'Source-based': source_based,

    'Keyword-based': keyword_based

})

plt.figure(figsize=(6,6))

plt.pie(
```

```
    reason_counts,
    labels=reason_counts.index,
    autopct='%1.1f%%',
    startangle=90
)
plt.title("Labeling Reason Distribution")
plt.show()
```

Prepare dataset for supervised learning

```
df_model = df[df['label'] != 'Unknown'].copy()
df_model['target'] = df_model['label'].map({'Fake': 0, 'Real': 1})
print("Final model dataset shape:", df_model.shape)
print(df_model['label'].value_counts())
X = df_model['clean_text']
y = df_model['target']
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
print("Train size:", len(X_train))
print("Test size:", len(X_test))
def plot_confusion_matrix(y_true, y_pred, title):
    cm = confusion_matrix(y_true, y_pred)
    plt.figure(figsize=(5,4))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
```

```
            xticklabels=['Fake', 'Real'],
            yticklabels=['Fake', 'Real'])
    plt.xlabel("Predicted")
    plt.ylabel("Actual")
    plt.title(title)
    plt.show()
def evaluate_model(name, y_true, y_pred, y_prob=None, train_time=None):
    result = {
        'Model': name,
        'Accuracy': accuracy_score(y_true, y_pred),
        'Precision': precision_score(y_true, y_pred),
        'Recall': recall_score(y_true, y_pred),
        'F1-score': f1_score(y_true, y_pred),
        'MCC': matthews_corrcoef(y_true, y_pred),
        'Train Time (s)': train_time
    }
    if y_prob is not None:
        result['ROC-AUC'] = roc_auc_score(y_true, y_prob)
        precision_vals, recall_vals, _ = precision_recall_curve(y_true, y_prob)
        result['PR-AUC'] = auc(recall_vals, precision_vals)
        result['Log Loss'] = log_loss(y_true, y_prob)
    else:
        result['ROC-AUC'] = np.nan
        result['PR-AUC'] = np.nan
        result['Log Loss'] = np.nan
    return result
```

Logistic Regression model

```
lr_pipeline = Pipeline([
    ('tfidf', TfidfVectorizer(max_features=10000, ngram_range=(1,2))),
    ('clf', LogisticRegression(max_iter=1000))
])
start_time = time.time()
lr_pipeline.fit(X_train, y_train)
lr_train_time = time.time() - start_time
y_pred_lr = lr_pipeline.predict(X_test)
y_prob_lr = lr_pipeline.predict_proba(X_test)[:, 1]
print("Logistic Regression")
print("Accuracy:", accuracy_score(y_test, y_pred_lr))
print(classification_report(y_test, y_pred_lr, target_names=['Fake', 'Real']))
plot_confusion_matrix(y_test, y_pred_lr, "Logistic Regression Confusion Matrix")
lr_result = evaluate_model(
    "Logistic Regression",
    y_test,
    y_pred_lr,
    y_prob=y_prob_lr,
    train_time=lr_train_time
)
```

XGBoost model

```
xgb_pipeline = Pipeline([
    ('tfidf', TfidfVectorizer(max_features=10000, ngram_range=(1,2))),
    ('clf', XGBClassifier(
```

```
        n_estimators=200,
        max_depth=6,
        learning_rate=0.1,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42,
        eval_metric='logloss'
    ))
])
start_time = time.time()
xgb_pipeline.fit(X_train, y_train)
xgb_train_time = time.time() - start_time
y_pred_xgb = xgb_pipeline.predict(X_test)
y_prob_xgb = xgb_pipeline.predict_proba(X_test)[:, 1]
print("XGBoost")
print("Accuracy:", accuracy_score(y_test, y_pred_xgb))
print(classification_report(y_test, y_pred_xgb, target_names=['Fake', 'Real']))
plot_confusion_matrix(y_test, y_pred_xgb, "XGBoost Confusion Matrix")
xgb_result = evaluate_model(
    "XGBoost",
    y_test,
    y_pred_xgb,
    y_prob=y_prob_xgb,
    train_time=xgb_train_time
)
```

BiLSTM Model

```
max_words = 10000
max_len = 200
tokenizer = Tokenizer(num_words=max_words, oov_token="<OOV>")
tokenizer.fit_on_texts(X_train)
X_train_seq = tokenizer.texts_to_sequences(X_train)
X_test_seq = tokenizer.texts_to_sequences(X_test)
X_train_pad = pad_sequences(X_train_seq, maxlen=max_len, padding='post', truncating='post')
X_test_pad = pad_sequences(X_test_seq, maxlen=max_len, padding='post', truncating='post')
dl_model = Sequential([
    Embedding(input_dim=max_words, output_dim=128, input_length=max_len),
    Bidirectional(LSTM(64)),
    Dropout(0.5),
    Dense(32, activation='relu'),
    Dropout(0.3),
    Dense(1, activation='sigmoid')
])
dl_model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
dl_model.summary()
early_stop = EarlyStopping(
    monitor='val_loss',
```

```
    patience=3,
    restore_best_weights=True
)
start_time = time.time()
history = dl_model.fit(
    X_train_pad,
    y_train,
    validation_split=0.2,
    epochs=10,
    batch_size=32,
    callbacks=[early_stop],
    verbose=1
)
dl_train_time = time.time() - start_time
dl_loss, dl_acc = dl_model.evaluate(X_test_pad, y_test, verbose=0)
y_prob_dl = dl_model.predict(X_test_pad).flatten()
y_pred_dl = (y_prob_dl > 0.5).astype(int)
print("BiLSTM Results")
print("Accuracy:", accuracy_score(y_test, y_pred_dl))
print(classification_report(y_test, y_pred_dl, target_names=['Fake', 'Real']))
plot_confusion_matrix(y_test, y_pred_dl, "BiLSTM Confusion Matrix")
dl_result = evaluate_model(
    "BiLSTM",
    y_test,
    y_pred_dl,
```

```
    y_prob=y_prob_dl,
    train_time=dl_train_time
)
```

DistilRoBERTa transformer Model

```
train_texts = X_train.tolist()
test_texts = X_test.tolist()
train_labels = y_train.tolist()
test_labels = y_test.tolist()
train_df = pd.DataFrame({
    'text': train_texts,
    'label': train_labels
})
test_df = pd.DataFrame({
    'text': test_texts,
    'label': test_labels
})
train_dataset = Dataset.from_pandas(train_df)
test_dataset = Dataset.from_pandas(test_df)
model_name = "distilroberta-base"
tokenizer_tr = AutoTokenizer.from_pretrained(model_name)
transformer_model = AutoModelForSequenceClassification.from_pretrained(model_name,
num_labels=2)
def tokenize_function(examples):
    return tokenizer_tr(
        examples["text"],
```

```
        padding="max_length",
        truncation=True,
        max_length=128
    )
train_dataset = train_dataset.map(tokenize_function, batched=True)
test_dataset = test_dataset.map(tokenize_function, batched=True)
train_dataset = train_dataset.remove_columns(["text"])
test_dataset = test_dataset.remove_columns(["text"])
train_dataset.set_format("torch")
test_dataset.set_format("torch")
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    precision = precision_score(labels, predictions)
    recall = recall_score(labels, predictions)
    f1 = f1_score(labels, predictions)
    accuracy = accuracy_score(labels, predictions)
    probs = torch.nn.functional.softmax(torch.tensor(logits), dim=1)[:, 1].numpy()
    roc_auc = roc_auc_score(labels, probs)
    return {
        "accuracy": accuracy,
        "precision": precision,
        "recall": recall,
        "f1": f1,
        "roc_auc": roc_auc
```

```
    }
training_args = TrainingArguments(
    output_dir="./distilroberta_results",
    eval_strategy="epoch",
    save_strategy="no",
    logging_strategy="epoch",
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=1,
    weight_decay=0.01,
    fp16=torch.cuda.is_available(),
    report_to="none"
) trainer = Trainer(
    model=transformer_model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)
start_time = time.time()
trainer.train()
transformer_train_time = time.time() - start_time
transformer_eval = trainer.predict(test_dataset)

transformer_logits = transformer_eval.predictions
```

```
y_pred_transformer = np.argmax(transformer_logits, axis=1)
y_prob_transformer = torch.nn.functional.softmax(
    torch.tensor(transformer_logits), dim=1
)[:, 1].numpy()
print("DistilRoBERTa Results")
print(classification_report(test_labels, y_pred_transformer, target_names=["Fake", "Real"]))
plot_confusion_matrix(test_labels, y_pred_transformer, "DistilRoBERTa Confusion Matrix")
transformer_result = {
    'Model': 'DistilRoBERTa',
    'Accuracy': accuracy_score(test_labels, y_pred_transformer),
    'Precision': precision_score(test_labels, y_pred_transformer),
    'Recall': recall_score(test_labels, y_pred_transformer),
    'F1-score': f1_score(test_labels, y_pred_transformer),
    'MCC': matthews_corrcoef(test_labels, y_pred_transformer),
    'ROC-AUC': roc_auc_score(test_labels, y_prob_transformer),
    'PR-AUC': auc(
        precision_recall_curve(test_labels, y_prob_transformer)[1],
        precision_recall_curve(test_labels, y_prob_transformer)[0]
    ),
    'Log Loss': log_loss(test_labels, y_prob_transformer),
    'Train Time (s)': transformer_train_time
}
```